



NOTEHERO NOTEHERO NOTEHERO

Physics





Addis Ababa Science & Technology University

Department of Physics

Lecture

General Physics (phys1001)

June, 2022

11/07/2019

Outline

- Uniform Circular Motion
- Radial and Tangential accelerations.
- Kinematics of uniform circular motion.
- problem-solving procedure (related to uniform circular motion)



Addis Ababa Science & Technology University

Department of Physics

Lecture

General Physics (phys1001)

June, 2022

11/07/2019



Addis Ababa Science & Technology University

Department of Physics

Lecture

General Physics (phys1001)

June, 2022

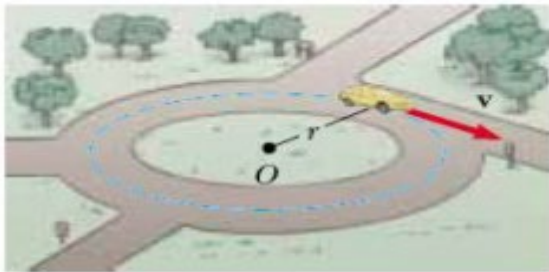
11/07/2019

Outline

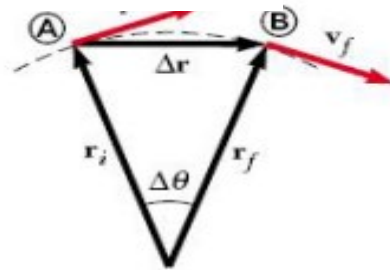
- Uniform Circular Motion
- Newton's law of Universal Gravitation
- Kepler's Laws, Satellites Motion & Weightlessness
- Work, Energy & Linear Momentum

Circular motion

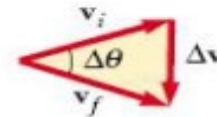
- Is a movement of an object along the circumference of a circle or rotation along a circular path.
- It can be **uniform** with constant speed or
- It can be **non-uniform** with change in rate of rotation.



(a)



(b)



(c)

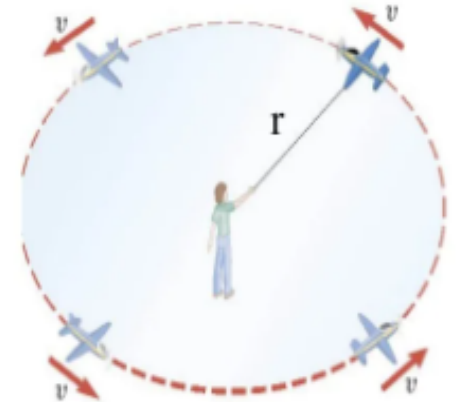
Figure 3.7: A car moving in a circular path with constant speed

Uniform Circular Motion

Uniform circular motion is the motion of an object traveling at a constant (uniform) speed on a circular path.

Period T is the time required to travel once around the circle, that is, to make one complete revolution.

$$T = \frac{2\pi r}{v}$$



3

$$\frac{|\Delta \mathbf{v}|}{v} = \frac{|\Delta \mathbf{r}|}{r}$$

$$|\bar{\mathbf{a}}| = \frac{|\Delta \mathbf{v}|}{\Delta t} = \frac{v}{r} \frac{|\Delta \mathbf{r}|}{\Delta t}$$

$$a_c = \frac{v^2}{r}$$

In case of uniform circular motion:

1. **Velocity:** is always **tangent to the circular path** and **perpendicular** to the **radius** of the circular path.
2. **Acceleration:** is always **perpendicular** to the circular **path**, and points towards the center of the circle.
 - Such acceleration is called the **centripetal acceleration**

Example 1.

What is the centripetal acceleration of the Earth as it moves in its orbit around the Sun?

$$\begin{aligned} a_c &= \frac{v^2}{r} = \frac{\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{4\pi^2 r}{T^2} \\ &= \frac{4\pi^2 (1.496 \times 10^{11} \text{ m})}{(1 \text{ yr})^2} \left(\frac{1 \text{ yr}}{3.156 \times 10^7 \text{ s}}\right)^2 \\ &= 5.93 \times 10^{-3} \text{ m/s}^2 \end{aligned}$$

Tangential and Radial Acceleration

As the particle moves along the curved path, the direction of the total acceleration vector $\mathbf{a}$ changes from point to point. This vector can be resolved into two components, based on an origin at the center of the dashed circle:

❖ Centripetal(radial)

- Toward center
- Due to changes in direction, perpendicular to $\mathbf{v}$
- $\mathbf{a}_c = \frac{v^2}{r}$, $\mathbf{a}_r = -\frac{v^2}{r} = -\mathbf{a}_c$

❖ Tangential (linear)

- Tangent to circle
- Due to changes in speed, parallel to $\mathbf{v}$

$$a_t = r\alpha = \frac{d|v|}{dt}$$

Total acceleration = the vector sum of the two component vectors:

$$\mathbf{a} = \mathbf{a}_t + \mathbf{a}_r$$

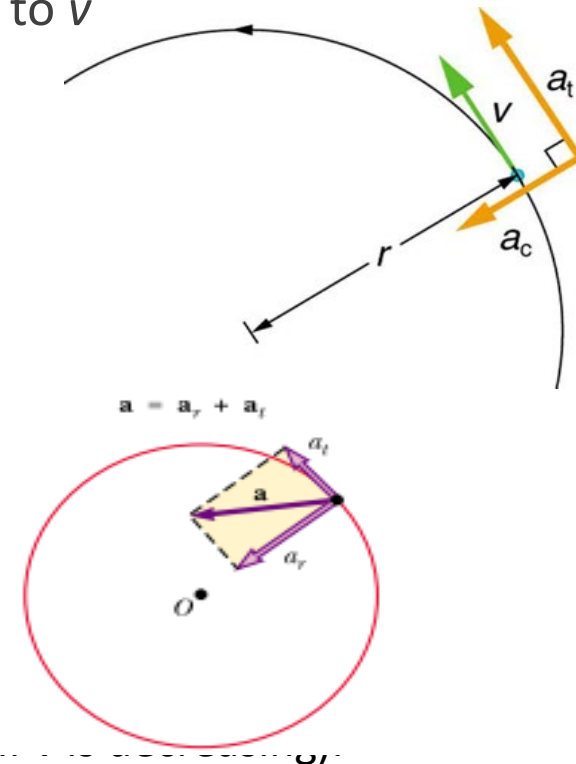
$$\mathbf{a} = \mathbf{a}_t + \mathbf{a}_r = \frac{d|\mathbf{v}|}{dt} \hat{\boldsymbol{\theta}} - \frac{v^2}{r} \hat{\mathbf{r}}$$

Magnitude of the total acceleration is:

$$a = \sqrt{a_r^2 + a_t^2}$$

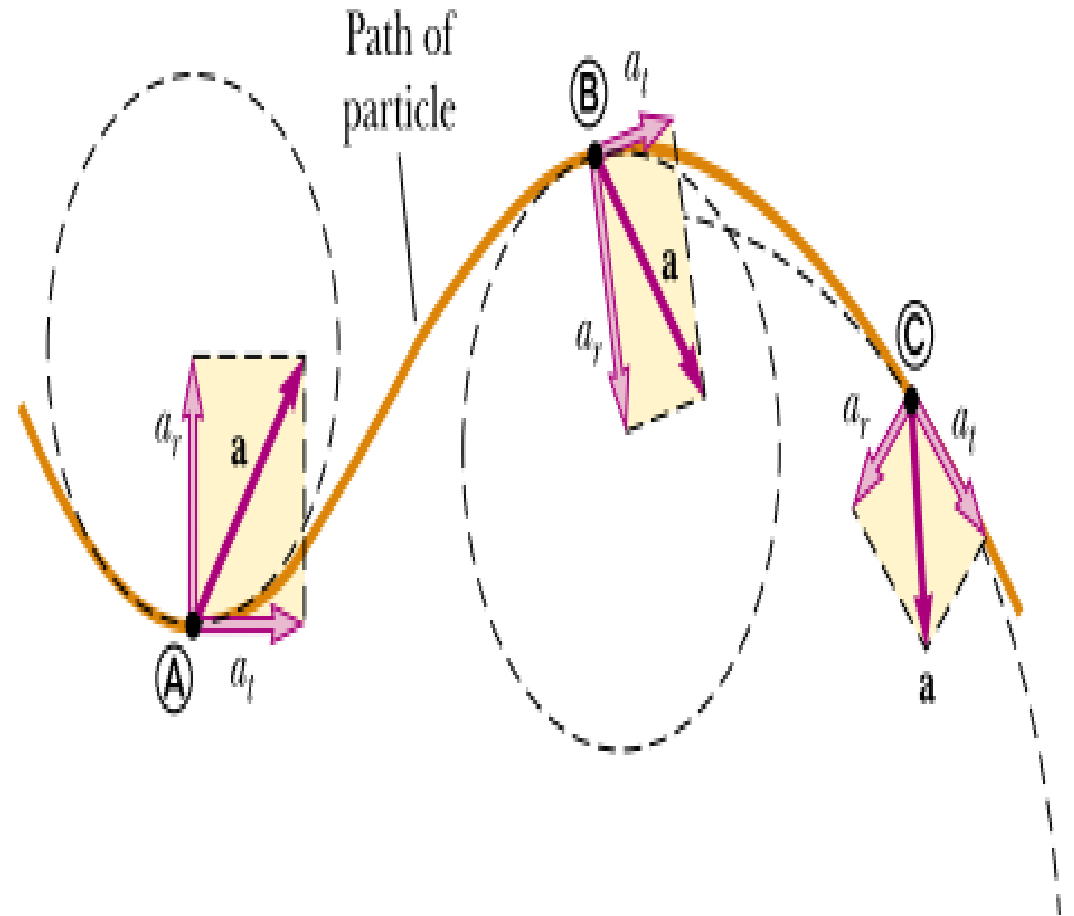
$$\text{direction, } \theta = \tan^{-1}\left(\frac{a_t}{a_r}\right)$$

N:B The direction of $\mathbf{a}_t$ is either in the same direction as $\mathbf{v}$ (if v is increasing) or opposite to $\mathbf{v}$,



Note:

- ✓ In uniform circular motion (speed, $v = \text{constant}$), the acceleration is always completely radial.
($a_t = 0$, $a_r \neq 0$).
- ✓ if the direction of $\mathbf{v}$ does not change, then there is no radial acceleration and the motion is one-dimensional
(i.e., $a_r = 0$, $a_t \neq 0$).



Kinematics of Rotational Motion

Rotational motion describes spinning motion

θ is like x

- $x = r\theta \rightarrow$ position

ω is like v

- $\omega = \frac{\Delta\theta}{\Delta t}$
- $v = r\omega \rightarrow$ velocity

α is like a

- $\alpha = \frac{\Delta\omega}{\Delta t}$
- $a_t = r\alpha \rightarrow$ acceleration

Equations of kinematics for rotational motion are same as for linear motion

$$\theta = \bar{\omega}t$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

Equations for centripetal force

From Newton's 2nd law of motion:

$$\Sigma F = ma$$

If a = centripetal acceleration

then ΣF = centripetal force

and so:

$$\begin{aligned} & \Sigma F = m v^2 / r \\ \text{and } & \Sigma F = m r \omega^2 \\ \text{and } & \Sigma F = m v \omega \end{aligned}$$

Example

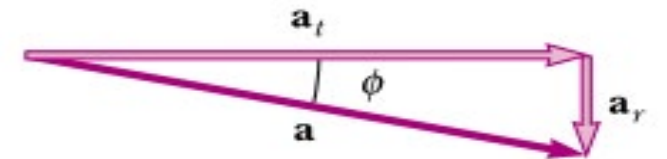
1. A car exhibits a constant acceleration of 0.300 m/s^2 parallel to the roadway. The car passes over a rise in the roadway such that the top of the rise is shaped like a circle of radius 500 m . At the moment the car is at the top of the rise, its velocity vector is horizontal and has a magnitude of 6.00 m/s . What is the direction of the total acceleration vector for the car at this instant?

Solution

Given: $a_t = 0.3 \text{ m/s}^2$, $V = 6 \text{ m/s}$. $R = 500 \text{ m}$

$$a_r = -\frac{v^2}{r} = -\frac{(6.00 \text{ m/s})^2}{500 \text{ m}} = -0.0720 \text{ m/s}^2$$

$$\begin{aligned} a &= \sqrt{a_r^2 + a_t^2} = \sqrt{(-0.0720)^2 + (0.300)^2} \text{ m/s}^2 \\ &= 0.309 \text{ m/s}^2 \end{aligned}$$



If ϕ is the angle between $\mathbf{a}$ and the horizontal, then

$$\phi = \tan^{-1} \frac{a_r}{a_t} = \tan^{-1} \left(\frac{-0.0720 \text{ m/s}^2}{0.300 \text{ m/s}^2} \right) = -13.5$$

This angle is measured downward from the horizontal.

Question 2

Calculate the maximum speed that a car of mass 800kg can go around a curve of radius 40m if the maximum frictional force available is 8kN.

The car will skid if the centripetal force required is greater than 8kN

$$\Sigma F = m v^2 / r$$

becomes: $v^2 = (\Sigma F \times r) / m$

$$= (8000\text{N} \times 40\text{m}) / (800\text{kg})$$

$$v^2 = 400$$

$$\text{maximum speed} = 20\text{ms}^{-1}$$

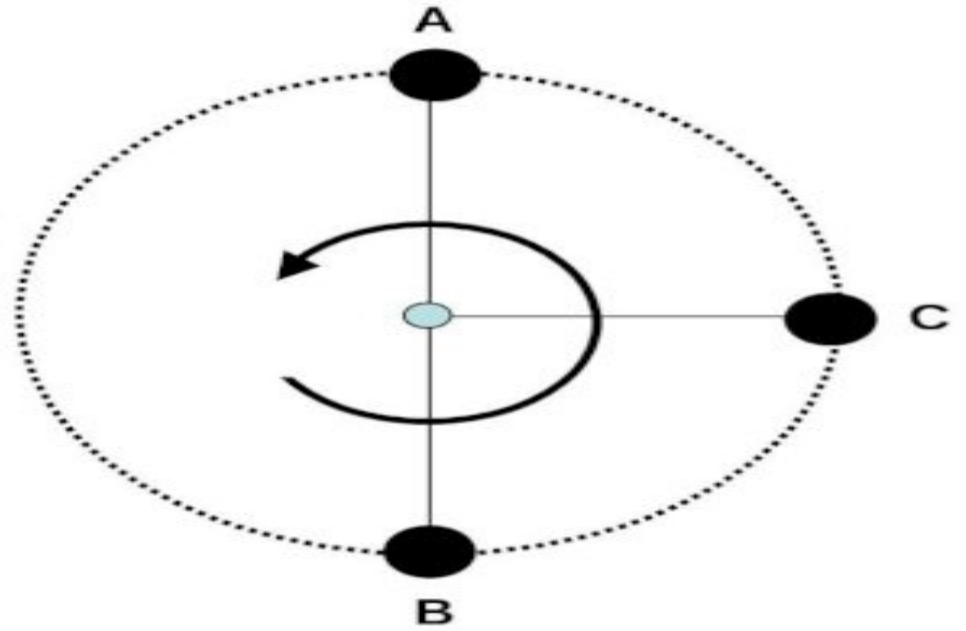
Question 3

A mass of 300g is whirled around a vertical circle using a piece of string of length 20cm at 3.0 revolutions per second.

Calculate the tension in the string at positions:

- (a) A – top
- (b) B – bottom and
- (c) C – string horizontal

The angular speed, $\omega = 3.0 \text{ rev s}^{-1}$
 $= 6\pi \text{ rad s}^{-1}$



Solution



(a) **A – top**

Both the weight of the mass and the tension in the string are pulling the mass towards the centre of the circle.

Therefore: $\Sigma F = mg + T$

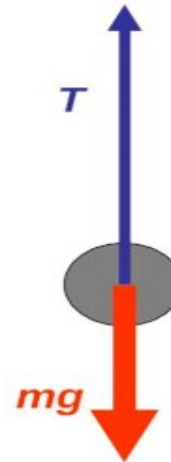
and so: $m r \omega^2 = mg + T$

giving: $T = m r \omega^2 - mg$

$$= [0.300\text{kg} \times 0.20\text{m} \times (6\pi \text{ rads}^{-1})^2] \\ - [0.300\text{kg} \times 9.8 \text{ Nkg}^{-1}]$$

$$= [21.32\text{N}] - [2.94\text{N}]$$

tension at A = 18.4N



(b) **B – bottom**

The weight is now acting away from the centre of the circle.

Therefore: $\Sigma F = T - mg$

and so: $m r \omega^2 = T - mg$

giving: $T = m r \omega^2 + mg$

$$= [0.300\text{kg} \times 0.20\text{m} \times (6\pi \text{ rads}^{-1})^2] \\ + [0.300\text{kg} \times 9.8 \text{ Nkg}^{-1}]$$

$$= [21.32\text{N}] + [2.94\text{N}]$$

tension at B = 24.3N



(c) **C – horizontal string**

The weight is acting perpendicular to the direction of the centre of the circle. It therefore has no effect on the centripetal force.

Therefore: $\Sigma F = T$

and so: $m r \omega^2 = T$

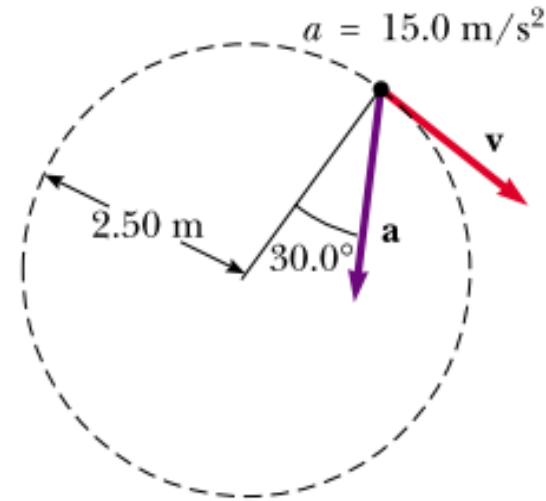
giving: $T = m r \omega^2$

$$= [0.300\text{kg} \times 0.20\text{m} \times (6\pi \text{ rads}^{-1})^2]$$

tension at C = 21.3N

Exercises

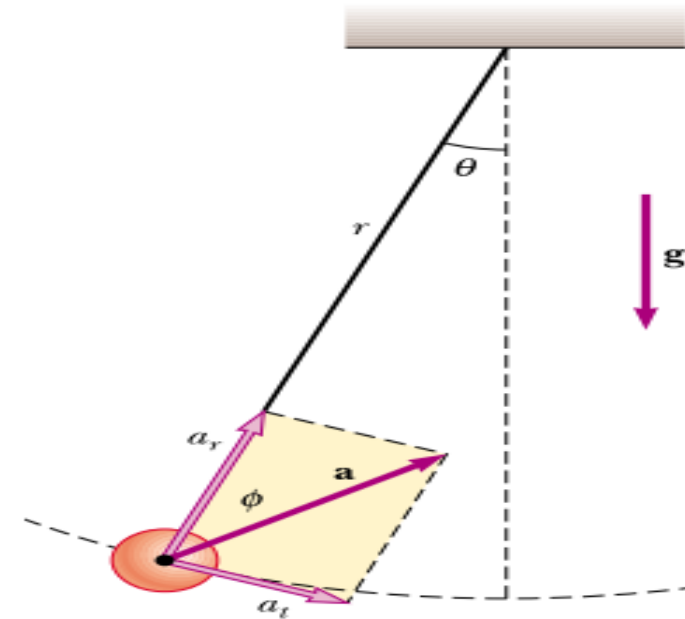
1. The Fig below represents the total acceleration of a particle moving clockwise in a circle of radius 2.50 m at a certain instant of time. At this instant, find
- (a) the radial acceleration,
 - (b) the speed of the particle, and
 - (c) its tangential acceleration.



2. An automobile whose speed is increasing at a rate of 0.600 m/s^2 travels along a circular road of radius 20.0 m. When the instantaneous speed of the automobile is 4.00 m/s, find
- (a) the tangential acceleration component,
 - (b) the centripetal acceleration component, and
 - (c) the magnitude and direction of the total acceleration.

Exercises

3. A pendulum with a cord of length $r = 1.00\text{ m}$ swings in vertical plane as shown in the Fig below. When the pendulum is in the two horizontal positions $\theta = 90^\circ$ and $\theta = 270^\circ$, its speed is 5.00 m/s . Find
- (a) the magnitude of the radial acceleration
 - (b) tangential acceleration for these positions.
 - (c) draw vector diagrams to determine the direction of the total acceleration for these two positions.
 - (d) Calculate the magnitude and direction of the total acceleration.



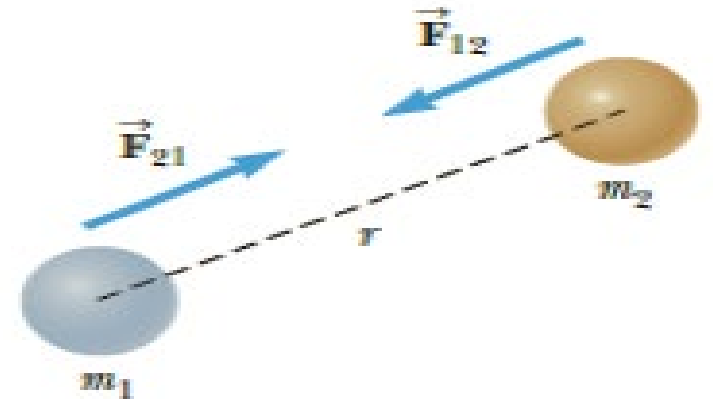
Newton's Law of Gravitation

- States that : “the Gravitational force, $\vec{F}_g$, between two particles with masses m_1 and m_2 and separated by a distance r , is directly proportional to the product of their masses and inversely proportional to the square of their separation. “
- the gravitational force is directed along the line joining the masses. That is;

$$\vec{F}_g = \frac{Gm_1m_2}{r^2}\hat{e}_r$$

where $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ is a constant of proportionality called the universal gravitational constant.

- The gravitational force is always attractive.

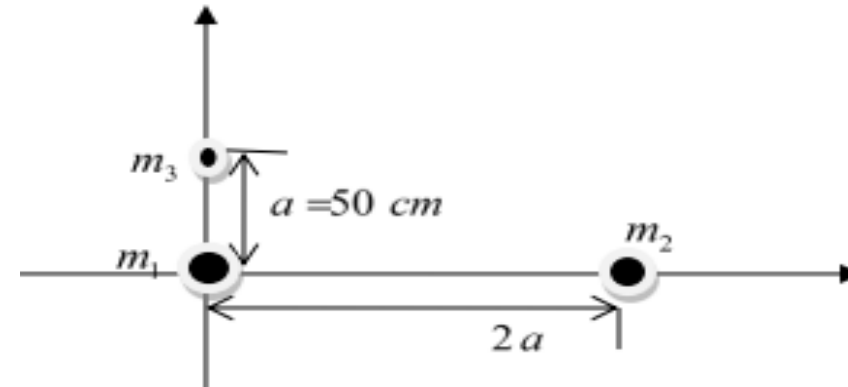


N: B F_{12} = the force on mass-1 by mass-2
 F_{21} = the force on mass-2 by mass-1

Example

1. Calculate the net gravitational force that mass $m_2 = 20 \text{ kg}$, $m_3 = 10 \text{ kg}$, exerts on mass $m_1 = 20 \text{ kg}$, for the case shown in Fig. below

Solution



$$\vec{F}_{12} = \frac{Gm_1m_2}{(2a)^2}\hat{i} = \frac{6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2} \times 20.0\text{kg} \times 20.0\text{kg}}{4 \times 0.25\text{m}^2} = 2.67 \times 10^{-8} \text{ N}\hat{i}$$

$$\vec{F}_{13} = \frac{Gm_1m_3}{(a)^2}\hat{j} = \frac{6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2} \times 10.0\text{kg} \times 20.0\text{kg}}{0.25\text{m}^2} = 5.34 \times 10^{-8} \text{ N}\hat{j}$$

The two forces are at right angle to each other. That is the force $\vec{F}_{12}$ directed along positive x-axis and $\vec{F}_{13}$ is directed along the positive-y-axis. The magnitude of resultant or the net force that the two masses exert on particle 1 is

$$F_1 = \sqrt{|\vec{F}_{12}|^2 + |\vec{F}_{13}|^2} = \sqrt{(2.67\text{N})^2 + (5.34\text{N})^2} = 5.97 \times 10^{-8} \text{ N}$$

The direction of the force is determined by calculating the angle θ that the resultant force makes with the positive x-axis and is given by $\theta = \tan^{-1} \left(\frac{5.34 \times 10^{-8}}{2.67 \times 10^{-8}} \right) = 63.4^\circ$.

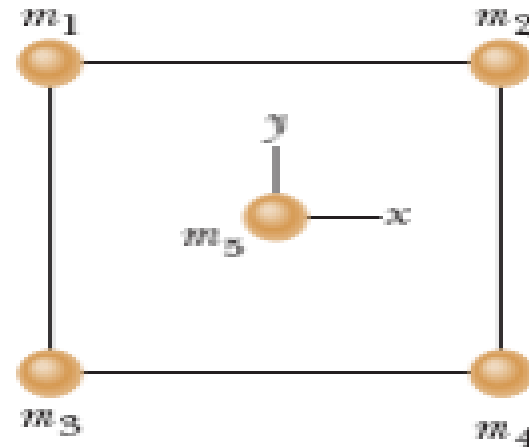
2. The Earth-Sun distance is about $1.5 \times 10^{11} \text{ m}$ = (one astronomical unit). The mass of the Earth is $6.0 \times 10^{24} \text{ kg}$ and the mass of the Sun is $2.0 \times 10^{30} \text{ kg}$. What is the magnitude of the force the Earth exerts on the Sun?

Solution:

$$\vec{F} = \frac{Gm_E m_S}{r^2} = \frac{6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2} \times 6.0 \times 10^{24} \text{ kg} \times 2.0 \times 10^{30} \text{ kg}}{(1.5 \times 10^{11} \text{ m})^2} = 3.56 \times 10^{21} \text{ N}$$

Exercise

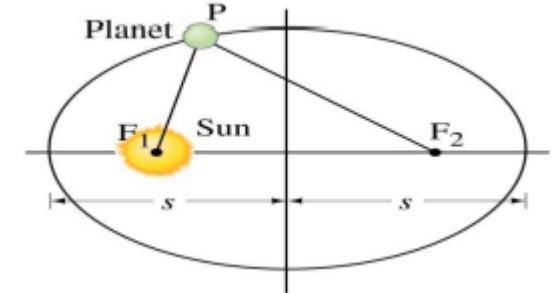
1. A square edge length 20.0 cm is formed by four spheres of masses $m_1 = 5 \text{ g}$, $m_2 = 3 \text{ g}$, $m_3 = 1 \text{ g}$ and $m_4 = 5 \text{ g}$ as shown in the Fig below. In unit-vector notation, what is the net gravitational force from them on a central sphere with mass $m_5 = 2.50 \text{ g}$?



Kepler's Law and the Motion of Planets

1. Kepler's First Law (Law of Orbits):

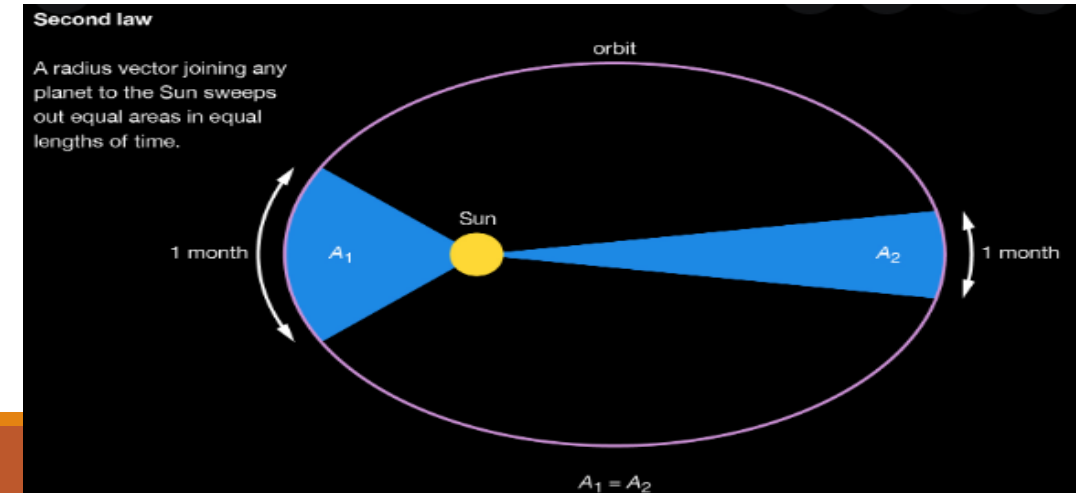
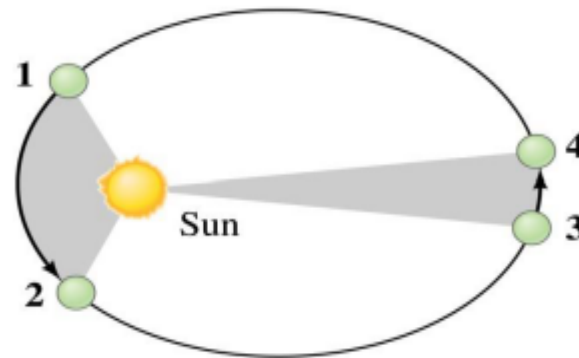
- States that "the orbit of each planet in the solar system is an **ellipse**, the Sun will be on one focus."



2. Kepler's Second Law (The Law of Areas)

- States that "the radius vector connecting the centers of the Sun and the Planet sweeps out equal areas in equal intervals of time."
- A planet moves fastest when it is closest to the sun and slowest when it is furthest from the sun.

$$v \propto \frac{1}{R}$$



3. Kepler's Third Law (Law of Period/Time/The Law of Harmony)

- States that “the square of the orbital period of a planet is proportional to the cube of the average distance between the centers of the planet and the sun.”

$$T^2 \propto r^3$$

$$or \quad T^2 \propto a^3$$

a = semi-major axis



$$\frac{T^2}{r^3} = \text{constant}$$

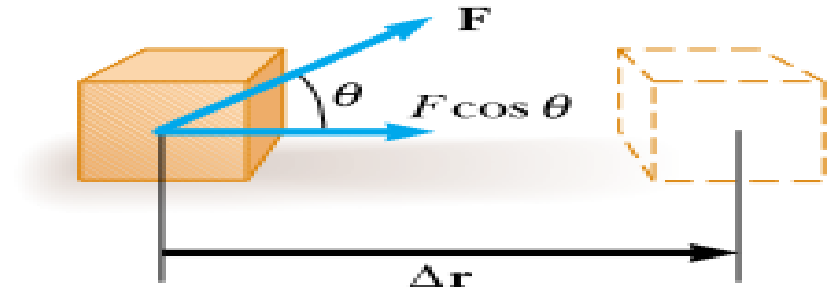
Planet	Period (s)	Average Distance (m)	T^2/R^3 (s ² /m ³)
Earth	3.156×10^7 s	1.4957×10^{11}	2.977×10^{-19}
Mars	5.93×10^7 s	2.278×10^{11}	2.975×10^{-19}

➤ Can be derived from Newton's law of gravitation

Work and Energy

The **work** W done on a system by an agent exerting a constant force on the system is the product of the magnitude F of the force, the magnitude Δr of the displacement of the point of application of the force, and $\cos \theta$, where θ is the angle between the force and displacement vectors:

$$W \equiv F \Delta r \cos \theta$$



Therefore for work to be done on an object, three essential conditions should be satisfied:

- Force must be exerted on the object
- The force must cause a motion or displacement
- The force should have a component along the line of displacement

Example:

1. A man cleaning a floor pulls a vacuum cleaner with a force of magnitude $F = 50.0 \text{ N}$ at an angle of 30.0° with the horizontal. Calculate the work done by the force on the vacuum cleaner as the vacuum cleaner is displaced 3.00 m to the right.

Solution:

$$W = F \Delta r \cos \theta = (50.0 \text{ N})(3.00 \text{ m})(\cos 30.0^\circ)$$

$$= 130 \text{ N}\cdot\text{m} = 130 \text{ J}$$

Example 2.

A particle moving in the xy plane undergoes a displacement $\Delta \mathbf{r} = (2.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}})$ m as a constant force $\mathbf{F} = (5.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}})$ N acts on the particle.

(A) Calculate the magnitudes of the displacement and the force.

Solution We use the Pythagorean theorem:

$$\Delta r = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(2.0)^2 + (3.0)^2} = 3.6 \text{ m}$$

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(5.0)^2 + (2.0)^2} = 5.4 \text{ N}$$

(B) Calculate the work done by $\mathbf{F}$.

Solution Substituting the expressions for $\mathbf{F}$ and $\Delta \mathbf{r}$ into Equation 7.3 and using Equations 7.4 and 7.5, we obtain

$$\begin{aligned} W &= \mathbf{F} \cdot \Delta \mathbf{r} = [(5.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}}) \text{ N}] \cdot [(2.0\hat{\mathbf{i}} + 3.0\hat{\mathbf{j}}) \text{ m}] \\ &= (5.0\hat{\mathbf{i}} \cdot 2.0\hat{\mathbf{i}} + 5.0\hat{\mathbf{i}} \cdot 3.0\hat{\mathbf{j}} + 2.0\hat{\mathbf{j}} \cdot 2.0\hat{\mathbf{i}} + 2.0\hat{\mathbf{j}} \cdot 3.0\hat{\mathbf{j}}) \text{ N} \cdot \text{m} \\ &= [10 + 0 + 0 + 6] \text{ N} \cdot \text{m} = 16 \text{ J} \end{aligned}$$

Work Done by a Spring

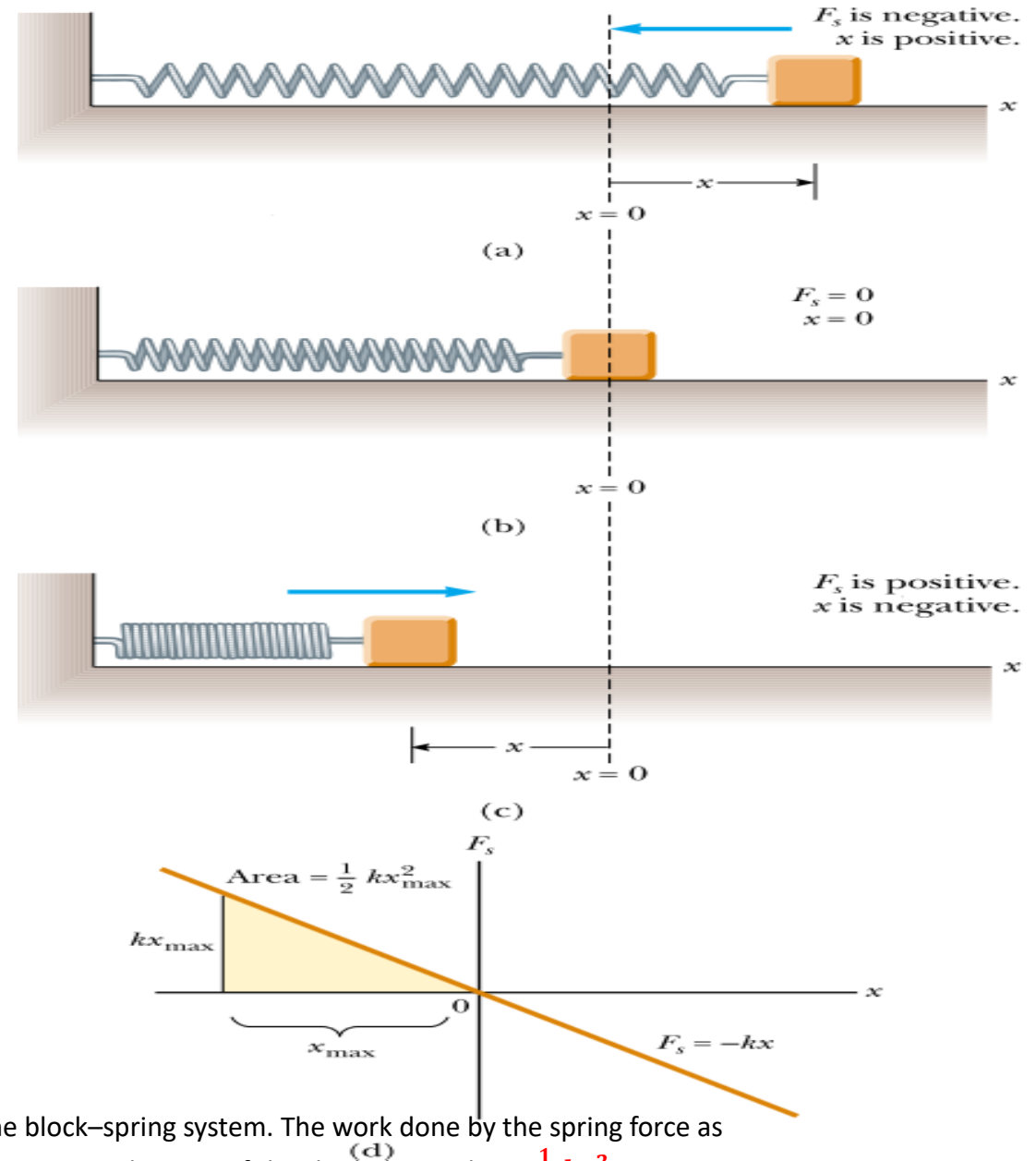
- A block on a horizontal, frictionless surface is connected to a spring. If the spring is either stretched or compressed a small distance from its unstretched (equilibrium) configuration, it exerts on the block a force that can be expressed as:

$$F_s = -kx \quad (\text{Hook's Law})$$

- The work done by the spring force on the block is :

$$W_s = \int_{x_i}^{x_f} F_s dx$$

$$W_s = \int_{x_i}^{x_f} (-kx) dx = \frac{1}{2} kx_i^2 - \frac{1}{2} kx_f^2$$



(d) Graph of F_s versus x for the block–spring system. The work done by the spring force as the block moves from $-x_{\max}$ to 0 is the area of the shaded triangle, $= \frac{1}{2} kx^2$

N:B The work done on the spring by an external agent (by an applied force on the spring)

- the applied force $\mathbf{F}_{app}$ is equal in magnitude and opposite in direction to the spring force $\mathbf{F}_s$:

$$F_{app} = -(-kx) = kx$$

- Now let us consider the work done on the spring by an external agent ($\mathbf{F}_{app}$) that stretches the spring very slowly from $x_i = 0$ to $x_f = x_{max}$

$$W_{F_{app}} = \int_0^{x_{max}} F_{app} dx = \int_0^{x_{max}} kx dx = \frac{1}{2} kx_{max}^2$$

- This work is equal to the negative of the work done by the spring force for this displacement.
- The work done by an applied force on a block–spring system between arbitrary positions of the block is:

$$W_{F_{app}} = \int_{x_i}^{x_f} F_{app} dx = \int_{x_i}^{x_f} kx dx = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$$

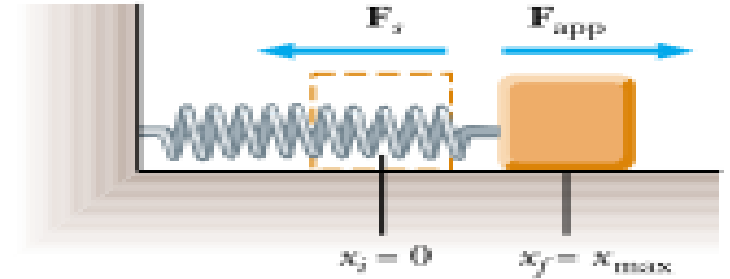


Figure 7.11 A block being pulled from $x_i = 0$ to $x_f = x_{max}$ on a frictionless surface by a force $\mathbf{F}_{app}$. If the process is carried out very slowly, the applied force is equal in magnitude and opposite in direction to the spring force at all times.

Notice that this is the negative of the work done by the spring. This is consistent with the fact that the spring force and the applied force are of equal magnitude but in opposite directions.

Energy:

- is defined as the capacity of a physical system to perform work.
- exists in several forms such kinetic, potential, thermal, chemical and other forms.
- its SI unit is joule (J).

Kinetic energy (KE)

- is the capacity of an object to do work by virtue of its motion.
- For an object of mass m and moving with speed v , the kinetic energy is calculated as:

$$KE = \frac{1}{2}mv^2$$

Potential Energy (PE) :

- is the energy that is stored in an object due to its position relative to some zero position.

Mathematically the potential energy is given by

$$\Delta U_g = mg\Delta y.$$

Work–Energy Theorem

- States that : In the case in which work is done on a system and the only change in the system is in its speed, then, the work done by the net force equals the change in kinetic energy of the system.

$$\sum W = K_f - K_i = \Delta K$$

Example-1

- A 6.0-kg block initially at rest is pulled to the right along a horizontal, frictionless surface by a constant horizontal force of 12 N. Find the speed of the block after it has moved 3.0 m.

Solution:

$$W = F x = (12 \text{ N})(3.0 \text{ m}) = 36 \text{ J}$$

$$W = K_f - K_i = \frac{1}{2}mv_f^2 - 0$$
$$v_f = \sqrt{\frac{2W}{m}} = \sqrt{\frac{2(36 \text{ J})}{6.0 \text{ kg}}} = 3.5 \text{ m/s}$$

Example2.

A 0.600-kg particle has a speed of 2.00 m/s at point A and kinetic energy of 7.50 J at point B. What is :

(a) its kinetic energy at A? (b) its speed at B? (c) the total work done on the particle as it moves from A to B?

Mechanical Energy (ME):

- Is defined as the sum of kinetic energy and potential energy.

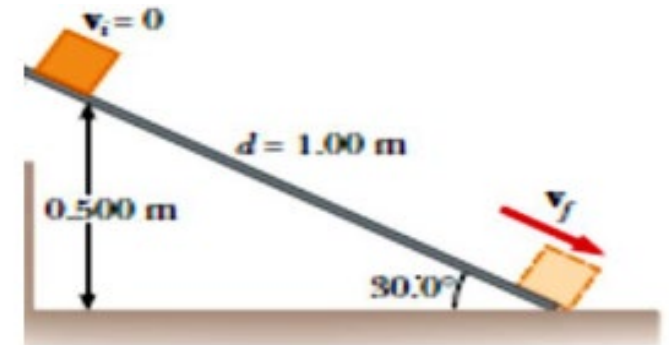
$$ME = KE + PE$$

- The mechanical energy of an isolated, friction-free system is conserved.

$$\frac{1}{2}mv_f^2 + mgy_f = \frac{1}{2}mv_i^2 + mgy_i$$

Example

1. A 3.00-kg crate slides down a ramp. The ramp is 1m in length and inclined at an angle of 30° , as shown in the fig. below. The crate starts from rest at the top, experiences a constant friction force of magnitude 5N, and continues to move a short distance on the horizontal floor after it leaves the ramp. Use energy methods to determine the speed of the crate at the bottom of the ramp.



Power

- is defined as the time rate of energy transfer.

$$P_{av} = \frac{W}{\Delta t} \qquad P = \frac{W}{\Delta t} = \vec{F} \cdot \frac{\Delta \vec{r}}{\Delta t} = \vec{F} \cdot \vec{v}$$

The SI unit of power is joules per second (J/s), also called the watt (W) (after James Watt):

$$1W = \frac{1J}{s} = 1kg \cdot \frac{m^2}{s^3}$$

A unit of power in the U.S. customary system is the horsepower (hp):

$$1hp = 746W$$

A unit of energy (or work) can now be defined in terms of the unit of power. One kilowatt-hour (kWh) is the energy transferred in 1 h at the constant rate of 1 kW = 1 000 J/s. The amount of energy represented by 1 kWh is:

$$1kWh = (10^3W)(3600s) = 3.6 \times 10^6J$$

Example

- An older model car of mass m accelerates from rest to speed v in 10 seconds. A newer, more powerful car of same mass accelerates from rest to $2v$ in the same time period. What is the ratio of the power of the newer car to that of the older car?

Sol. $W = \text{change in KE.}$ $P = W/t$

Linear Momentum

Momentum :

- is defined as the quality of a moving object to exert a force on anything that tries to stop it. The linear momentum of a particle or an object that can be modeled as a particle of mass m moving with a velocity $\mathbf{v}$ is defined to be the product of its mass and velocity:

$$\mathbf{P} = m\mathbf{v}$$

- is a vector quantity in the direction of the velocity, SI unit = kgm/s.

Using Newton's second law of motion, we can relate the linear momentum of a particle to the resultant force acting on the particle.

$$\vec{F} = m \frac{\Delta \vec{v}}{\Delta t} \Rightarrow \vec{F} = \frac{\Delta (m\vec{v})}{\Delta t} = \frac{\Delta \vec{P}}{\Delta t}$$

Therefore, the time rate of change of the linear momentum of a particle is equal to the net force acting on the particle.

$$\Delta \vec{P} = \vec{F} \Delta t$$

$$\vec{I}(\text{Impulse}) = \Delta \vec{P} = \vec{P}_f - \vec{P}_i = \vec{F} \Delta t$$

Conservation of Linear Momentum

- states that whenever two or more particles in an isolated system interact, the total momentum of the system remains constant.

$$\Rightarrow m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f}$$

$$\vec{P}_{1i} + \vec{P}_{2i} = \vec{P}_{1f} + \vec{P}_{2f} \quad \text{-----}$$

$$\vec{P}_{tot} = \vec{P}_1 + \vec{P}_2 = \text{constant}$$

(law of conservation of linear momentum)

Example:

A 60-kg archer stands at rest on frictionless ice and fires a 0.50-kg arrow horizontally at 50 m/s. With what velocity does the archer move across the ice after firing the arrow?

Solution:

Let m_1 be the mass of the archer and m_2 be mass of the arrow

If we consider the archer+ arrow as isolated system

$$\vec{P}_{tot} = \text{constant} \rightarrow \vec{P}_i = \vec{P}_f$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$v_{1i} = v_{2i} = 0$, Because initially the archer + arrow is at rest

$$0 = 60 \text{ kg } v_{1f} + (0.5 \text{ kg})(50 \text{ m/s})$$

$$v_{1f} = \frac{-25}{60} = -0.42 \text{ m/s}$$

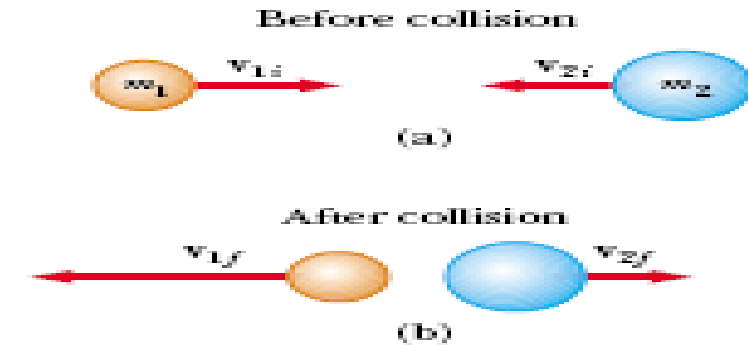
Collisions

- Depending on whether kinetic energy is conserved or not, collision can be classified into two:

1. Elastic collision

- both kinetic energy as well as total momentum of the system is conserved.

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$
$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$



2. Inelastic collision

- Only momentum is conserved
- Kinetic energy is not conserved

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

Perfectly Inelastic Collisions

- The two masses stick together after collision
- the two masses will have the same final velocity.

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f \quad v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

Note: It is important to indicate the direction of the velocities. We shall indicate *the velocity* as positive if a particle moves to the right, and negative if it moves to the left.

Two-Dimensional Collisions

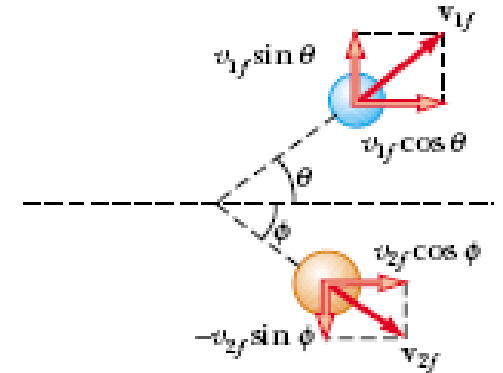
- For such two dimensional elastic collisions, we obtain two component equations for conservation of momentum

$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$



(a) Before the collision



(b) After the collision

• **Figure 9.13** An elastic glancing collision between two particles.

- Similarly, for such two dimensional elastic collisions, we obtain two component equations for conservation of kinetic energy

PROBLEM-SOLVING HINTS

Two-Dimensional Collisions

The following procedure is recommended when dealing with problems involving two-dimensional collisions between two objects:

- Set up a coordinate system and define your velocities with respect to that system. It is usually convenient to have the x axis coincide with one of the initial velocities.
- In your sketch of the coordinate system, draw and label all velocity vectors and include all the given information.
- Write expressions for the x and y components of the momentum of each object before and after the collision. Remember to include the appropriate signs for the components of the velocity vectors.
- Write expressions for the total momentum of the system in the x direction before and after the collision and equate the two. Repeat this procedure for the total momentum of the system in the y direction.
- If the collision is *inelastic*, kinetic energy of the system is *not* conserved, and additional information is probably required. If the collision is *perfectly* inelastic, the final velocities of the two objects are equal. Solve the momentum equations for the unknown quantities.
- If the collision is *elastic*, kinetic energy of the system is conserved, and you can equate the total kinetic energy before the collision to the total kinetic energy after the collision to obtain an additional relationship between the velocities.

The Center of Mass

$$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \cdots + m_n x_n}{m_1 + m_2 + m_3 + \cdots + m_n} = \frac{\sum_i m_i x_i}{\sum_i m_i} = \frac{\sum_i m_i x_i}{M}$$

$$y_{\text{CM}} = \frac{\sum_i m_i y_i}{M} \quad \text{and} \quad z_{\text{CM}} = \frac{\sum_i m_i z_i}{M}$$

$$\mathbf{r}_{\text{CM}} = x_{\text{CM}} \hat{\mathbf{i}} + y_{\text{CM}} \hat{\mathbf{j}} + z_{\text{CM}} \hat{\mathbf{k}} = \frac{\sum_i m_i x_i \hat{\mathbf{i}} + \sum_i m_i y_i \hat{\mathbf{j}} + \sum_i m_i z_i \hat{\mathbf{k}}}{M}$$
$$\mathbf{r}_{\text{CM}} = \frac{\sum_i m_i \mathbf{r}_i}{M}$$

where $\mathbf{r}_i$ is the position vector of the i th particle, defined by

$$\mathbf{r}_i = x_i \hat{\mathbf{i}} + y_i \hat{\mathbf{j}} + z_i \hat{\mathbf{k}}$$

-----The end ----



Addis Ababa Science & Technology University

Department of Physics

General Physics (phys1001)

Chapter - 3
Fluid Mechanics

June, 2022

FLUID MECHANICS

- **Fluid mechanics**: is the branch of physics concerned with the mechanics of fluids in motion (fluid dynamics) or at rest (fluid statics) and the forces on them.
- Fluid: a fluid is a substance that continually deforms (flows) under an applied shear stress, or external force.
- Fluids are a phase of matter and include liquids and gases.

Properties of Bulk Matter

Definitions:

- **Elastic materials** : are materials that regain their original shape and size when the deforming force is removed.
- **Elastic deformation**: is a reversible deformation by a force applied within the elastic limit.
 - Beyond elastic limit a force applied on an object causes permanent and irreversible deformation called **Plastic deformation**.
- **Plastics materials**: do not regain their original shape and size when the deforming force is removed.

The elastic properties of solid materials are described in terms of **stress and strain**.

Stress and strain

A. Stress : is the force per unit area that is causing some deformation on an object

$$\text{Stress} = \frac{F}{A}$$

SI unit = $\frac{N}{m^2}$, called the Pascal (Pa), the same as the unit of pressure.

B. Strain:

- Is the change in configuration of a body divided by its initial configuration.
- measures the amount of deformation by the applied stress
- SI unit = unit less

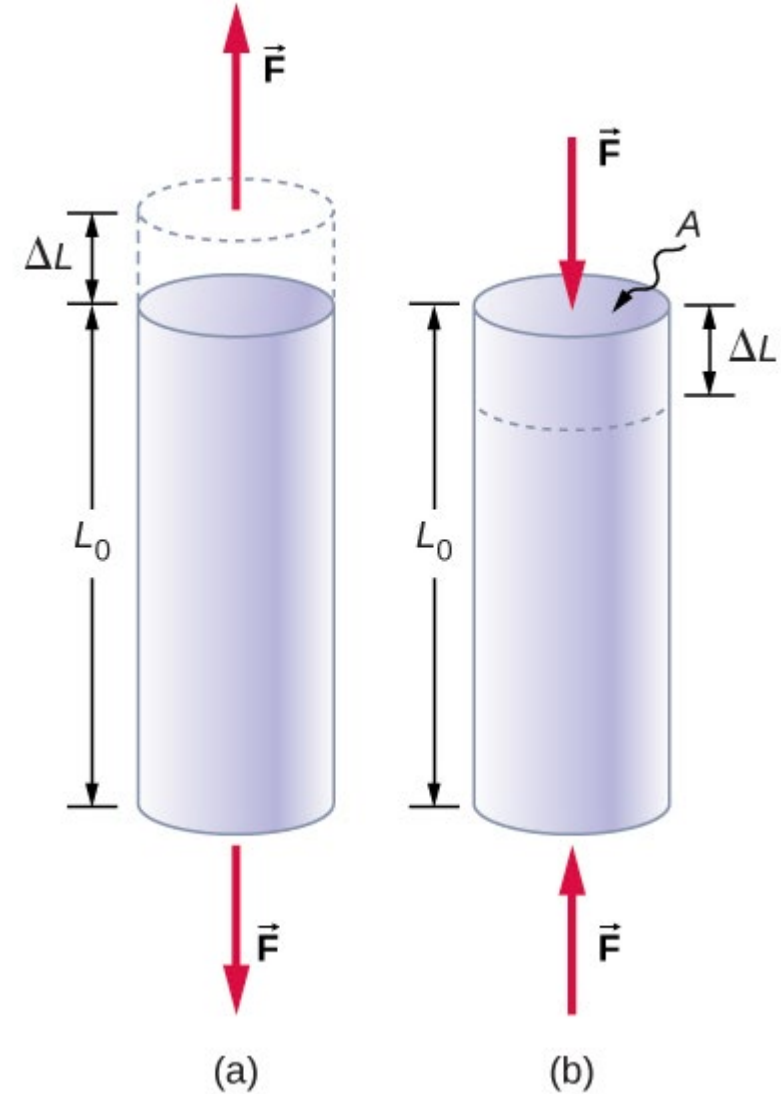
$$\text{Strain} = \frac{\text{Change in configuration}}{\text{Initial configuration}}$$

- Strain is given as a fractional change in either length (under tensile stress) or volume (under bulk stress) or geometry (under shear stress)

a) Tension: The rod is elongated by ΔL .

b) Compression: The rod is contracted by ΔL .

In both cases, the deforming force acts along the length of the rod and perpendicular to its cross section



kinds of Stress/ Strains:

1. Tensile Stress / Tensile Strain

- Tensile Stress:

$$\text{tensile stress} = \frac{F_{\perp}}{A}$$

- tensile strain : the fractional change in length of an object by a given tensile stress.
 - is a strain due to tensile stress

$$\text{tensile strain} = \frac{\Delta L}{L_0}.$$

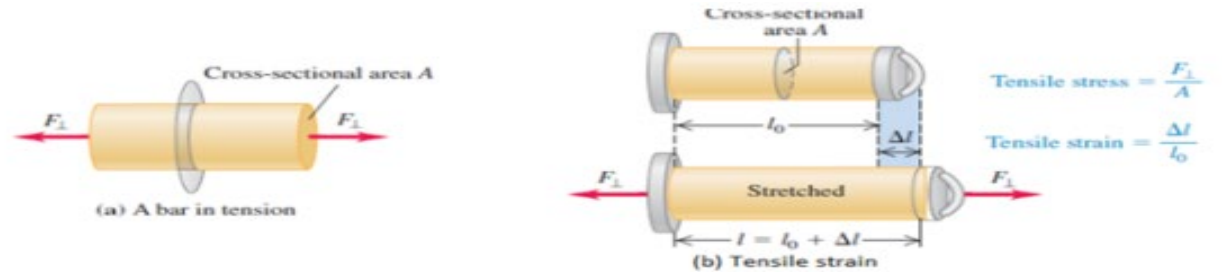


Figure : Shows a bar's a) tensile stress and b) tensile strain

$$\text{Tensile strain} = \frac{\Delta l}{l_0}.$$

2. shear stress /shear strain.

- Strain caused by shear stress is called **shear strain**

$$\text{Shear stress} = \frac{\Delta F}{A}$$

- The shear strain is defined as the ratio x/h , where x is the horizontal distance that the sheared face moves and h is the height of the object.

$$\text{Shear strain} = \frac{x}{h} = \tan \phi$$

3. bulk(volume) stress / bulk (volume) strain ,

- Volume Stress is a stress which causes volume deformation on an object and defined as the ratio of the change in the magnitude of the applied force ΔF to the surface area A .

$$\text{Volume stress} = \Delta p$$

- strain under bulk stress is called **bulk strain** (or **volume strain**), is the fractional change in volume. i.e the change in volume, ΔV , divided by the original volume V_0 :

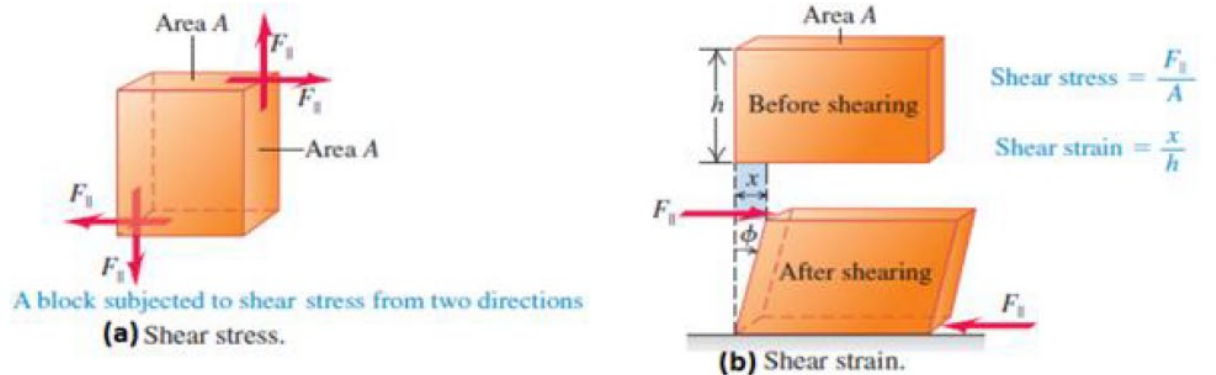


Figure 3.2: Shows an object deformed by a shear stress, a) shear stress and b) shear strain

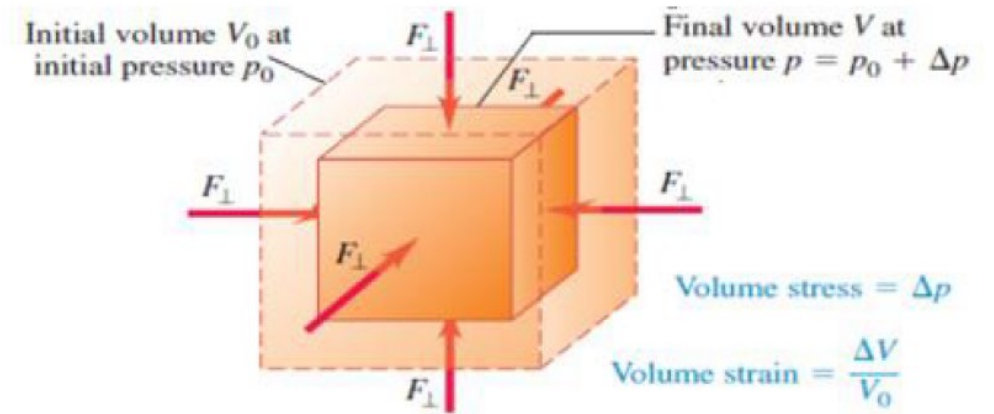


Figure 3.3: A block undergoing volume strain by the applied volume stress

$$\text{Volume strain} = \frac{\Delta V}{V_0}$$

Elastic modulus:

- The greater the stress, the greater the strain; however, the relation between strain and stress does not need to be linear.
- Only when stress is sufficiently low is the deformation it causes directly proportion to the stress value.
- The proportionality constant in this relation is called the **elastic modulus**.

In the linear limit of low stress values, the general relation between stress and strain is given by:

$$\text{stress} = (\text{elastic modulus}) \times \text{strain}.$$

analogous to Hooke's law ($F = -kx$),

❖ Corresponding to the three types of strains, there are three types of elastic module.

1. Young's Modulus:(Y)

- is the ratio of the tensile stress to the tensile strain.
- It measures the resistance of a solid to a change in its length and typically used to characterize a rod or wire stressed under either tension or compression.

$$Y = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{F_{\perp}/A}{\Delta l/l_0}.$$

Unit = force per unit area = N/m² = Pascal

2. Shear Modulus (S):

- is the ratio of shear stress to shear strain.
- is the measure of the resistance to motion of the planes within a solid parallel to each other
- Unit = Pascal,

$$S = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F_{\parallel}/A}{x/h}$$

3. Bulk Modulus (B)

- is the ratio of the volume stress to the volume strain.
- measures the resistance of solids or liquids to changes in their volume.
- SI unit = Pascal

$$B = \frac{\text{Volume stress}}{\text{Volume strain}} = \frac{-\Delta F/A}{\Delta V/V_0} = \frac{-\Delta P}{\Delta V/V_0}$$

Note: - Negative sign implies an increase in pressure (positive ΔP) causes a decrease in volume (negative ΔV) and vice versa.
- The reciprocal of the bulk modulus is called the [compressibility of the material](#)

Example:

1. Suppose that the tension in the cable is 940 N as the actor reaches the lowest point. What diameter should a 10-m-long steel wire have if we do not want it to stretch more than 0.5 cm under these conditions?

Substance	Young's Modulus (N/m ²)	Shear Modulus (N/m ²)	Bulk Modulus (N/m ²)
Tungsten	35×10^{10}	14×10^{10}	20×10^{10}
Steel	20×10^{10}	8.4×10^{10}	6×10^{10}
Copper	11×10^{10}	4.2×10^{10}	14×10^{10}
Brass	9.1×10^{10}	3.5×10^{10}	6.1×10^{10}
Aluminum	7.0×10^{10}	2.5×10^{10}	7.0×10^{10}

Solution

$$Y = \frac{F_{\perp}/A}{\Delta l/l_0} \Rightarrow A = \frac{Fl_0}{Y\Delta l} \quad A = \pi r^2 \quad d = 2r$$

Activity: A solid brass sphere is initially surrounded by air, and the air pressure exerted on it is $1.0 \times 10^5 \text{ N/m}^2$ (normal atmospheric pressure). The sphere is lowered into the ocean to a depth where the pressure is $2.0 \times 10^7 \text{ N/m}^2$. The volume of the sphere in air is 0.50 m^3 . By how much does this volume change once the sphere is submerged?

Density and Pressure in Static Fluids

- **Density (ρ)** : is the quantity of mass (m) per unit volume (V) of a body

$$\rho = \frac{m}{V}$$

- **Specific gravity (SG)**: is the ratio of the density of the substance to the density of another substance which is taken as a standard(water.)

Example:

A solid sphere made of wood has a radius of 0.1 m. The mass of the sphere is 1.0 kg. Determine a) density and b) specific gravity of the wood.

Solution

The volume of the sphere wood, $V = \frac{4}{3}\pi r^3 = \frac{4}{3} \times 3.14 \times (0.1)^3 = 4.18 \times 10^{-3} m^3$.

a) The density of the wood, $\rho = \frac{m}{V} = \frac{1.0 \text{ kg}}{4.18 \times 10^{-3} m^3} = 239 \text{ kg/m}^3$.

b) The specific gravity of the wood, $SG = \frac{\text{density of wood}}{\text{density of water}} = \frac{239 \text{ kg/m}^3}{1000 \text{ kg/m}^3} = 0.239$

Pressure

- is the ratio of the force acting perpendicular to surface to the surface area (A) on which the force acts.

$$P = \frac{F_{\perp}}{A}$$

- SI unit of pressure = N/m² called Pascal (Pa).

Variation of Pressure with Depth

- consider a liquid of density ρ at rest as shown in Fig. contained within an imaginary cylinder of cross-sectional area A extending from depth d to depth $d + h$.

$$\sum \mathbf{F} = PA\hat{\mathbf{j}} - P_0A\hat{\mathbf{j}} - Mg\hat{\mathbf{j}} = 0$$

$$PA - P_0A - \rho Ahg = 0$$

$$PA - P_0A = \rho Ahg$$

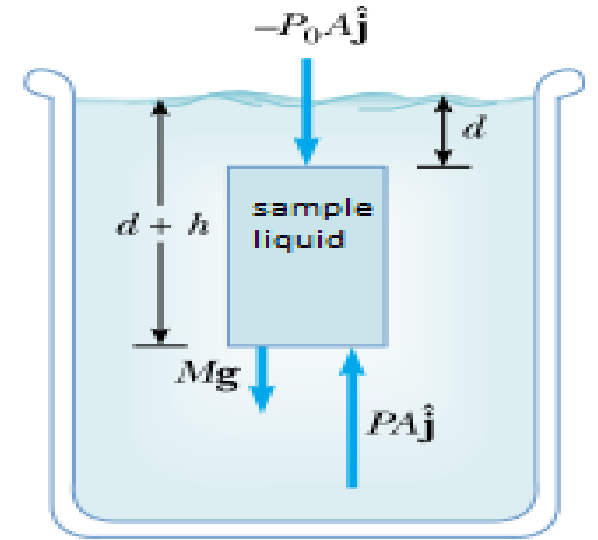
$$P = P_0 + \rho gh$$

(this is, the pressure P at a depth h below a point in the liquid)

$$P_0 = 1.00 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

- The pressure produced by the column of fluid of height h and density ρ is given by:

$$P_{\text{fluid}} = \rho gh$$

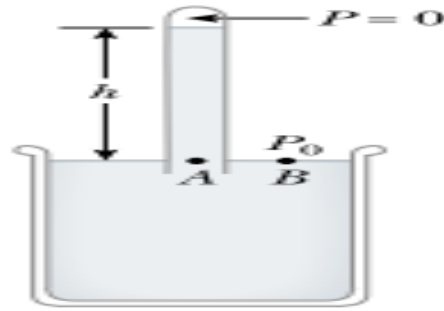


Note that:

- All points at same level in a fluid have same pressure.
- Fluid pressure increases with increase in the depth of the fluid.
- Fluid pressure does not depend on the shape of the container.

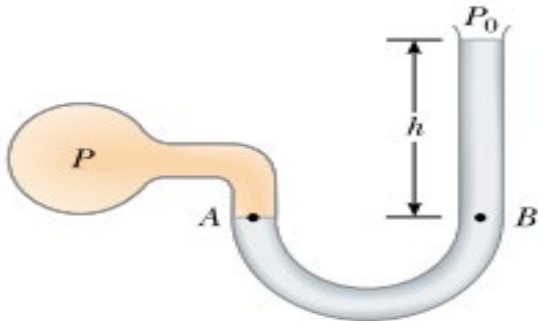
Pressure Measurements

1. A **mercury Barometer** : used to measure atmospheric pressure



$$P_0 = \rho_{\text{Hg}}gh \rightarrow h = \frac{P_0}{\rho_{\text{Hg}}g} = \frac{1.013 \times 10^5 \text{ Pa}}{(13.6 \times 10^3 \text{ kg/m}^3)(9.80 \text{ m/s}^2)} = 0.760 \text{ m}$$

2. **Open-tube Manometer**: used to measure the pressure of a gas contained in a vessel



$$P = P_0 + \rho gh.$$

The pressure P is called the **absolute pressure**.

- The difference $P - P_0$ is called the **Gauge pressure**.

Atmospheric Pressure: is the pressure due to the weight of the atmosphere exerted on the surface of the Earth. Atmospheric pressure decreases with increase in altitude as a result of decrease in the density of the air.

Gauge pressure: is the difference in pressure between a system and the surrounding atmosphere.

$$P_{gauge} = P_{system} - P_{atmosphere}$$

Absolute Pressure:

- Also called the total pressure, is the sum of gauge pressure and atmospheric pressure:

$$P_{absolute} = P_{gauge} + P_{atmospheric}$$

In most cases the absolute pressure in fluids cannot be negative. Fluids push rather than pull, so the smallest absolute pressure is zero. (A negative absolute pressure is a pull.) Thus the smallest possible gauge pressure is $P_{gauge} = -P_{atmosphere}$

Example

1. A submerged wreck is located 18.3 m beneath the surface of the ocean. Determine
a) gauge pressure and b) absolute pressure .

Note: the density of sea water = 1025 kg/m³

$$\text{a)} \quad P_{gauge} = \rho gh = (1025 \text{ kg/m}^3)(9.80 \text{ m/s}^2)(18.3 \text{ m}) = 1.83 \times \frac{10^5 \text{ N}}{\text{m}^2} = 182 \text{ kPa}.$$

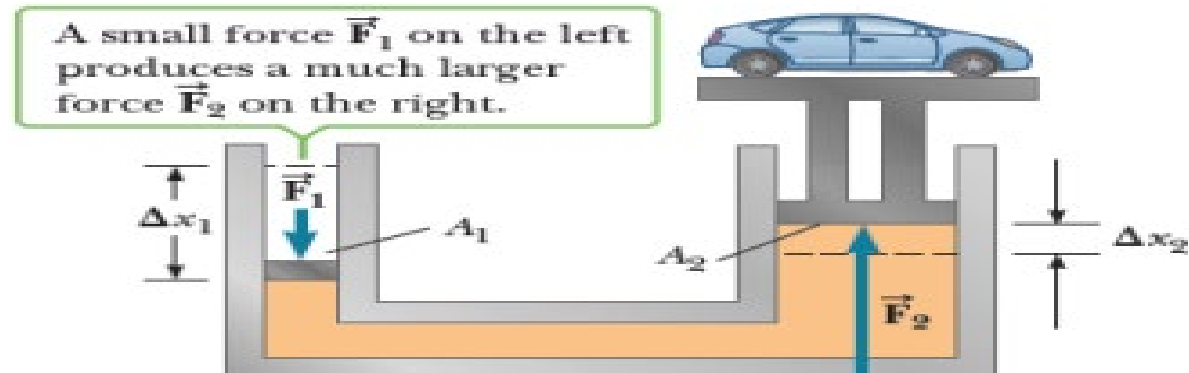
$$\text{b)} \quad P_{abs} = P_{gau} + P_{atm} = (1.83 \times 10^5 + 1.013 \times 10^5) \text{ N/m}^2 = 2.84 \times 10^5 \text{ N/m}^2 = 2.84 \text{ kPa} .$$

Pascal's Principle

- states that pressure applied to a confined fluid in a container is transmitted equally to all regions of the fluid and to the walls of the container.

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$



Example:

The small piston of a hydraulic lift has an area of 0.20 m^2 . A car weighing $1.2 \times 10^4 \text{ N}$ sits on a rack mounted on the large piston. The large piston has an area of 0.90 m^2 . How large force must be applied to the small piston to support the car?

Given:

$$A_1 = 0.20 \text{ m}^2$$

$$A_2 = 0.90 \text{ m}^2$$

$$F_2 = 1.2 \times 10^4 \text{ N}$$

Unknown:

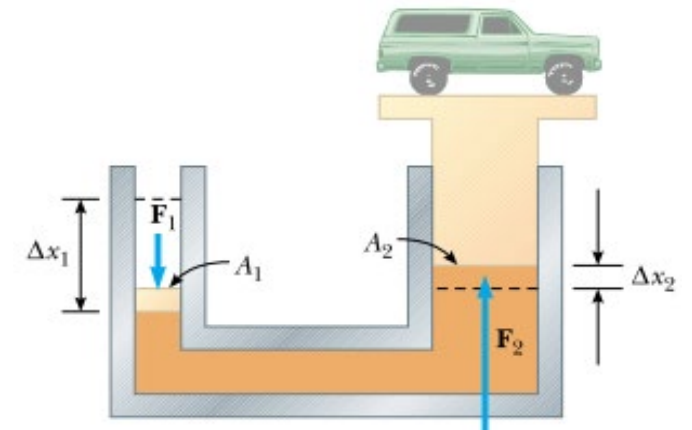
$$F_1 = ?$$

Solution:

$$F_1/A_1 = F_2/A_2 \Rightarrow F_1 = F_2 \left(A_1/A_2 \right)$$

$$= (1.2 \times 10^4 \text{ N}) \left(\frac{0.20 \text{ m}^2}{0.90 \text{ m}^2} \right)$$

$$= 2.7 \times 10^3 \text{ N}$$



Example2.(Pascal Principle)

In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 5.00 cm. This pressure is transmitted by a liquid to a piston that has a radius of 15.0 cm. What force must the compressed air exert to lift a car weighing 13 300 N? What air pressure produces this force?

Solution Because the pressure exerted by the compressed air is transmitted undiminished throughout the liquid, we have

$$F_1 = \left(\frac{A_1}{A_2} \right) F_2 = \frac{\pi(5.00 \times 10^{-2} \text{ m})^2}{\pi(15.0 \times 10^{-2} \text{ m})^2} (1.33 \times 10^4 \text{ N})$$

$$= 1.48 \times 10^3 \text{ N}$$

The air pressure that produces this force is

$$P = \frac{F_1}{A_1} = \frac{1.48 \times 10^3 \text{ N}}{\pi(5.00 \times 10^{-2} \text{ m})^2}$$
$$= 1.88 \times 10^5 \text{ Pa}$$

This pressure is approximately twice atmospheric pressure.

Archimedes's Principle

- States that the magnitude of the buoyant force is always equal to the weight of the fluid displaced by the object.

$$F_b = W_{dis.fluid}$$

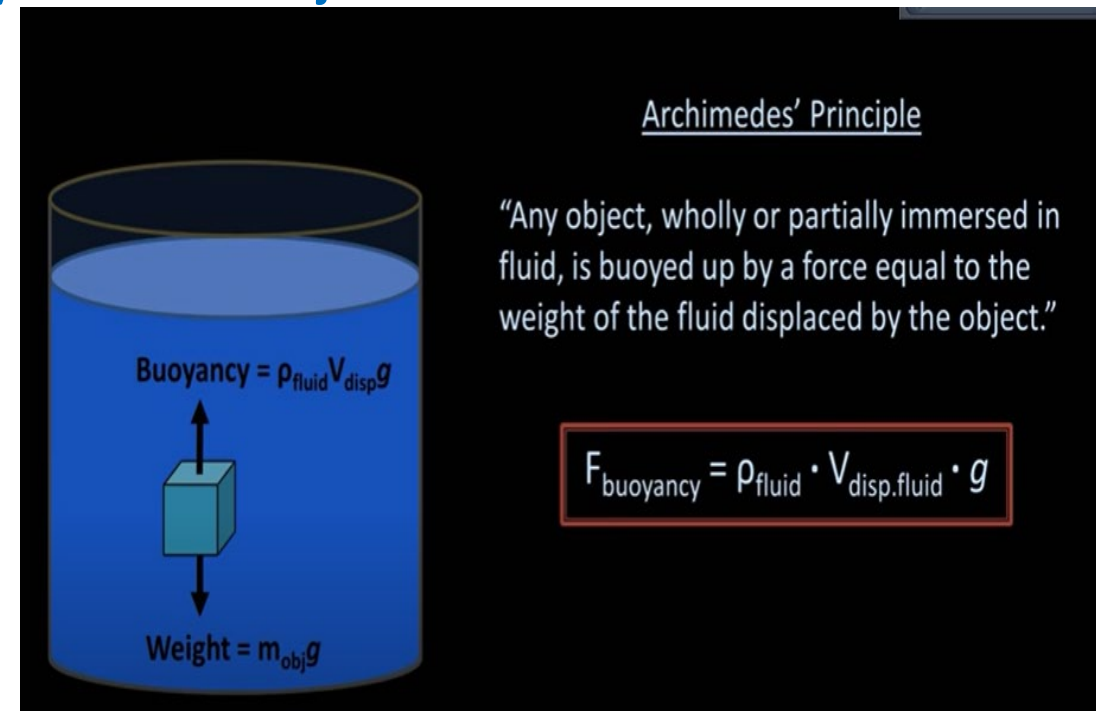
- Buoyant force:** the upward force exerted by a fluid on any immersed object.

$$B = (P_b - P_t)A = (\rho_{fluid}gh)A = \rho_{fluid}gV_{dis.fl}$$

Case 1: Totally Submerged Object :

$$B = \rho_{fluid}gV_{dis.fl} = \rho_{fluid}gV_{obj},$$

- $\rho_{obj} < \rho_{fluid}$, the object accelerates upward(floats).
- $\rho_{obj} > \rho_{fluid}$, the object accelerates downward(Sinks)
- $\rho_{obj} = \rho_{fluid}$, the F_{net} on the object is zero and it remains in **Equilibrium** (Floats.)



Case 2. Partially submerged (Floating Object)

- $\rho_{obj} < \rho_{fluid}$, in static equilibrium floating on the surface of a fluid.
- In this case, the upward buoyant force is balanced by the downward gravitational force acting on the object.

$$\frac{V_{dis.fluid}}{V_{obj(in)}} = \frac{\rho_{obj}}{\rho_{fluid}}$$

- Thus, this equation tells us that the fraction of the volume of a floating object, i.e., below the fluid surface is equal to the ratio of the density of the object to that of the fluid.

N:B. Buoyant force can be also defined as:
 $B = \text{Weight in air} - \text{Weight in fluid}$

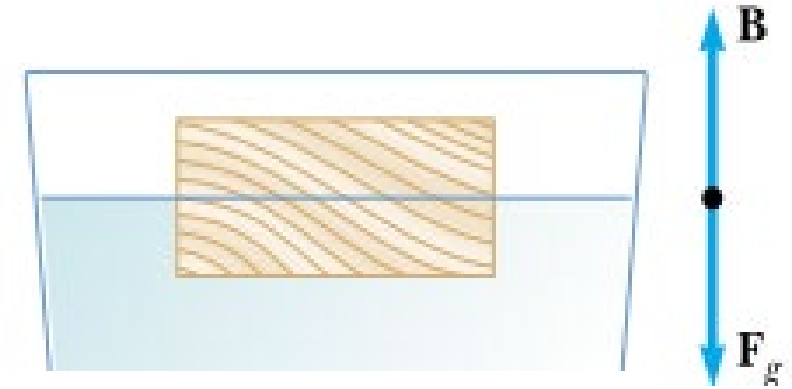
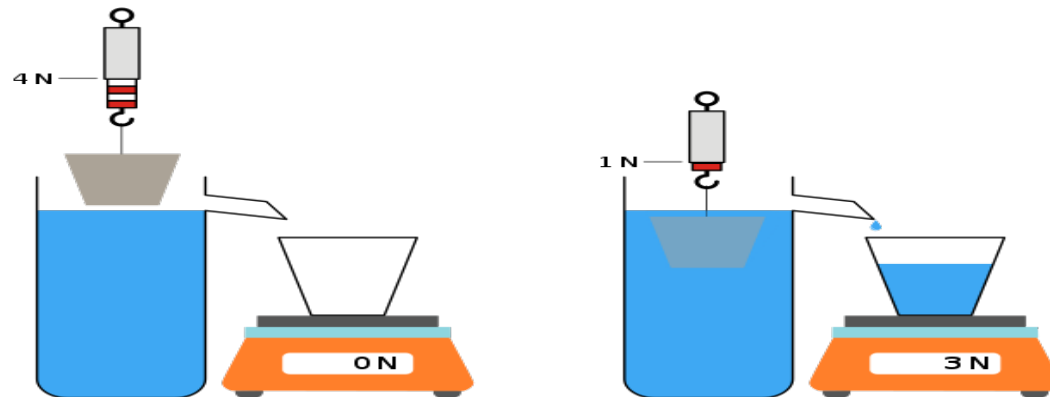
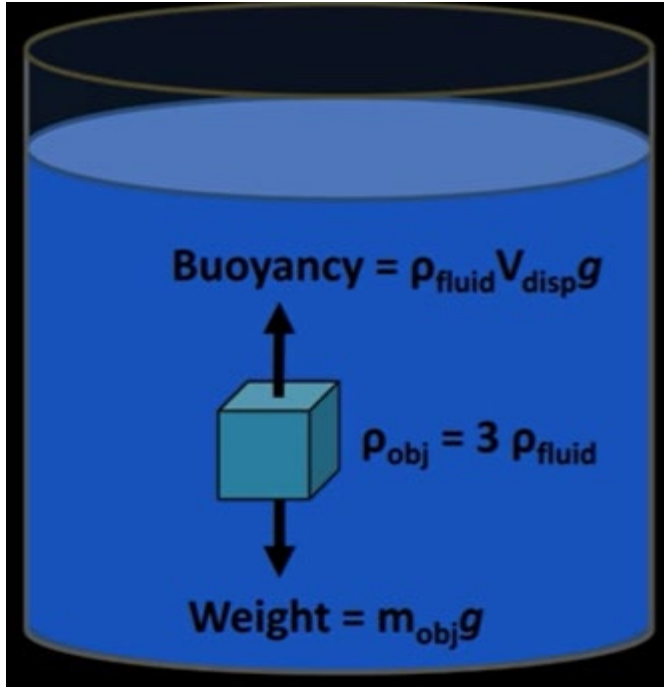


Fig. An object floating on the surface of a fluid experiences two forces, the gravitational force F_g and the buoyant force B . Because the object floats in equilibrium, $B = F_g$.



Example

1. What acceleration will a totally submerged object experience if its density is three times that of the fluid in which it is submerged?



$$\begin{aligned}\Sigma F &= m a \\ F_{\text{weight}} - F_{\text{buoyancy}} &= m_{\text{obj}} a_{\text{obj}} \\ (m_{\text{obj}} g) - (\rho_{\text{fluid}} V_{\text{obj}} g) &= m_{\text{obj}} a_{\text{obj}} \\ (\cancel{3\rho_{\text{fluid}}} \cancel{V_{\text{obj}}} g) - (\cancel{\rho_{\text{fluid}}} \cancel{V_{\text{obj}}} g) &= (\cancel{3\rho_{\text{fluid}}} \cancel{V_{\text{obj}}}) a_{\text{obj}} \\ 3g - g &= 3a_{\text{obj}} \\ a_{\text{obj}} &= \frac{2}{3}g = \frac{2}{3}(9.8 \text{ m/s}^2) = 6.5 \text{ m/s}^2\end{aligned}$$

2. It's obvious that a piece of steel will sink in water, yet ships made of steel floats.

Example

3. Suppose a 60.0-kg woman floats in fresh water with 97.0% of her volume submerged when her lungs are full of air. What is her average density?

We can find the woman's density by solving the equation

$$\text{fraction submerged} = \frac{\rho_{\text{obj}}}{\rho_{\text{fl}}}$$

for the density of the object. This yields

$$\rho_{\text{obj}} = \rho_{\text{person}} = (\text{fraction submerged}) \cdot \rho_{\text{fl}}.$$

We know both the fraction submerged and the density of water, so we can calculate the woman's density.

Solution

Entering the known values into the expression for her density, we obtain

$$\rho_{\text{person}} = 0.970 \cdot (10^3 \frac{\text{kg}}{\text{m}^3}) = 970 \frac{\text{kg}}{\text{m}^3}.$$

Note: The fraction submerged is the ratio of the volume submerged to the volume of the object, or

$$\text{fraction submerged} = \frac{V_{\text{sub}}}{V_{\text{obj}}} = \frac{V_{\text{fl}}}{V_{\text{obj}}}.$$

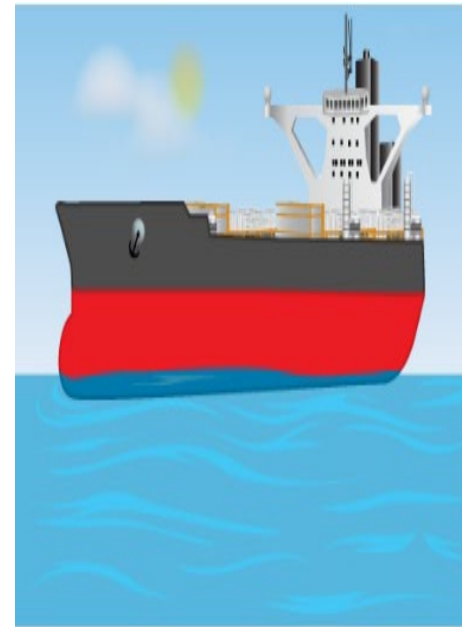
The volume submerged equals the volume of fluid displaced, which we call V_{fl} . Now we can obtain the relationship between the densities by substituting $\rho = \frac{m}{V}$ into the expression. This gives

$$\frac{V_{\text{fl}}}{V_{\text{obj}}} = \frac{m_{\text{fl}}/\rho_{\text{fl}}}{m_{\text{obj}}/\rho_{\text{obj}}},$$

where ρ_{obj} is the average density of the object and ρ_{fl} is the density of the fluid. Since the object floats, its mass and that of the displaced fluid are equal, so they cancel from the equation, leaving

$$\text{fraction submerged} = \frac{\rho_{\text{obj}}}{\rho_{\text{fl}}}.$$

We can use this relationship to measure densities.



(a)



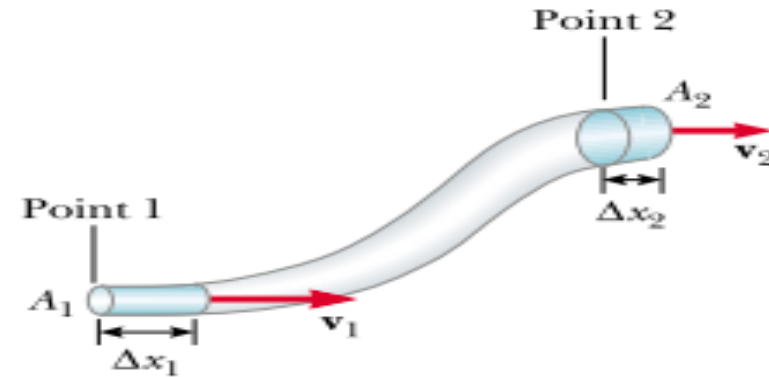
(b)

Figure 14.22 An unloaded ship (a) floats higher in the water than a loaded ship (b).

Equation of Continuity

- It states that the product of the area and the fluid speed at all points along a pipe is constant for an incompressible fluid.

$$A_1 v_1 = A_2 v_2 = \text{constant}$$



Note:

- the speed (v) is inversely proportional to Area (A)
- The product Av is called volume flux or flow rate which has the dimensions of volume per unit time

Example (14.8)

1. A water hose 2.50 cm in diameter is used by a gardener to fill a 30.0-L bucket. The gardener notes that it takes 1.00 min to fill the bucket. A nozzle with an opening of cross-sectional area 0.500 cm² is then attached to the hose. The nozzle is held so that water is projected horizontally from a point 1.00 m above the ground. Over what horizontal distance can the water be projected?

Solution We identify point 1 within the hose and point 2 at the exit of the nozzle. We first find the speed of the water in the hose from the bucket-filling information. The cross-sectional area of the hose is

$$A_1 = \pi r^2 = \pi \frac{d^2}{4} = \pi \left(\frac{(2.50 \text{ cm})^2}{4} \right) = 4.91 \text{ cm}^2$$

According to the data given, the volume flow rate is equal to 30.0 L/min:

$$A_1 v_1 = 30.0 \text{ L/min} = \frac{30.0 \times 10^3 \text{ cm}^3}{60.0 \text{ s}} = 500 \text{ cm}^3/\text{s}$$
$$v_1 = \frac{500 \text{ cm}^3/\text{s}}{A_1} = \frac{500 \text{ cm}^3/\text{s}}{4.91 \text{ cm}^2} = 102 \text{ cm/s} = 1.02 \text{ m/s}$$

Now we use the continuity equation for fluids to find the speed $v_2 = v_{xi}$ with which the water exits the nozzle. The subscript i anticipates that this will be the *initial* velocity

component of the water projected from the hose, and the subscript x recognizes that the initial velocity vector of the projected water is in the horizontal direction.

$$A_1 v_1 = A_2 v_2 = A_2 v_{xi} \longrightarrow v_{xi} = \frac{A_1}{A_2} v_1$$
$$v_{xi} = \frac{4.91 \text{ cm}^2}{0.500 \text{ cm}^2} (1.02 \text{ m/s})$$
$$= 10.0 \text{ m/s}$$

We now shift our thinking away from fluids and to projectile motion because the water is in free fall once it exits the nozzle. A particle of the water falls through a vertical distance of 1.00 m starting from rest, and lands on the ground at a time that we find from Equation 2.12:

$$y_f = y_i + v_{yi}t - \frac{1}{2}gt^2$$
$$-1.00 \text{ m} = 0 + 0 - \frac{1}{2}(9.80 \text{ m/s}^2)t^2$$
$$t = \sqrt{\frac{2(1.00 \text{ m})}{9.80 \text{ m/s}^2}} = 0.452 \text{ s}$$

In the horizontal direction, we apply Equation 2.12 with $a_x = 0$ to a particle of water to find the horizontal distance:

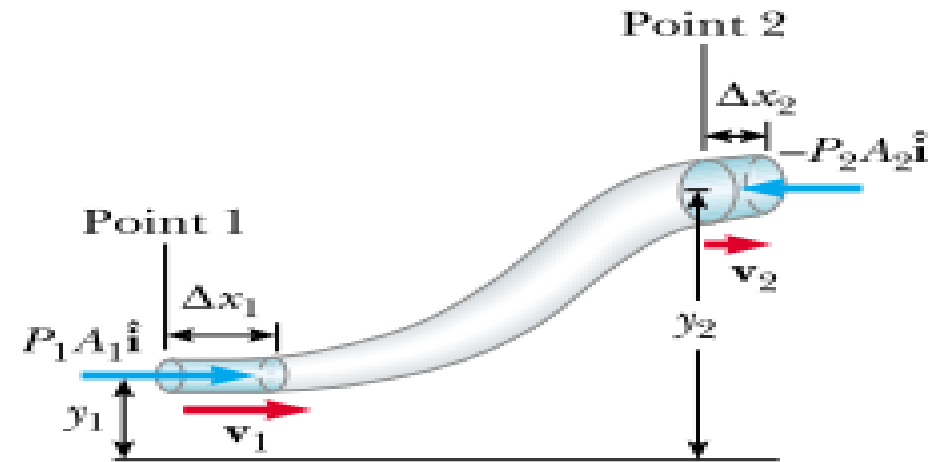
$$x_f = x_i + v_{xi}t = 0 + (10.0 \text{ m/s})(0.452 \text{ s}) = 4.52 \text{ m}$$

Bernoulli's Equation

States that:

$$P + \frac{1}{2}\rho v^2 + \rho gy = \text{constant}$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gy_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gy_2$$



BERNOULLI'S EQUATION

$$P + \frac{\rho}{2}V^2 + \rho gh = \text{constant}$$

ENERGY FORM

$$\frac{P}{\rho} + \frac{V^2}{2} + gh = \text{constant} \quad (\text{along a streamline})$$

PRESSURE FORM

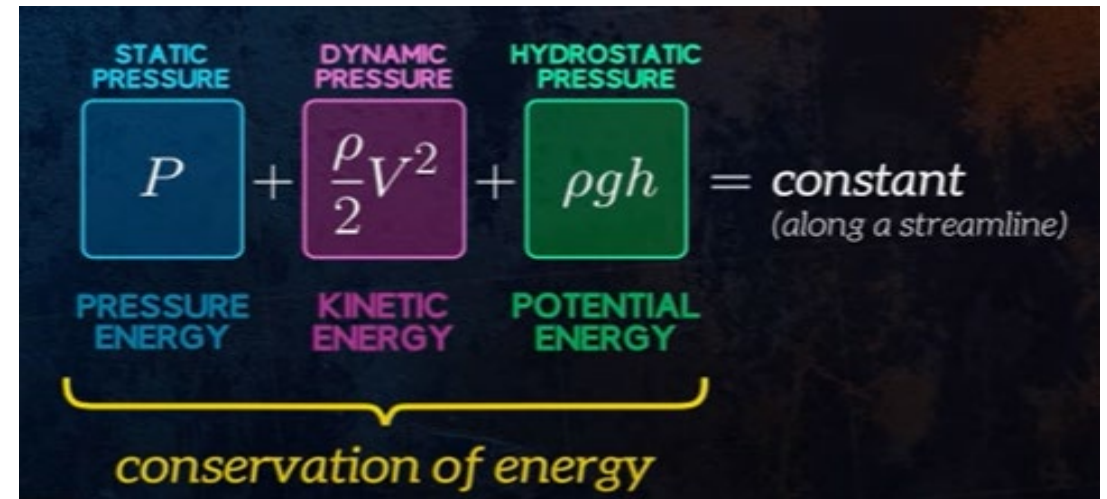
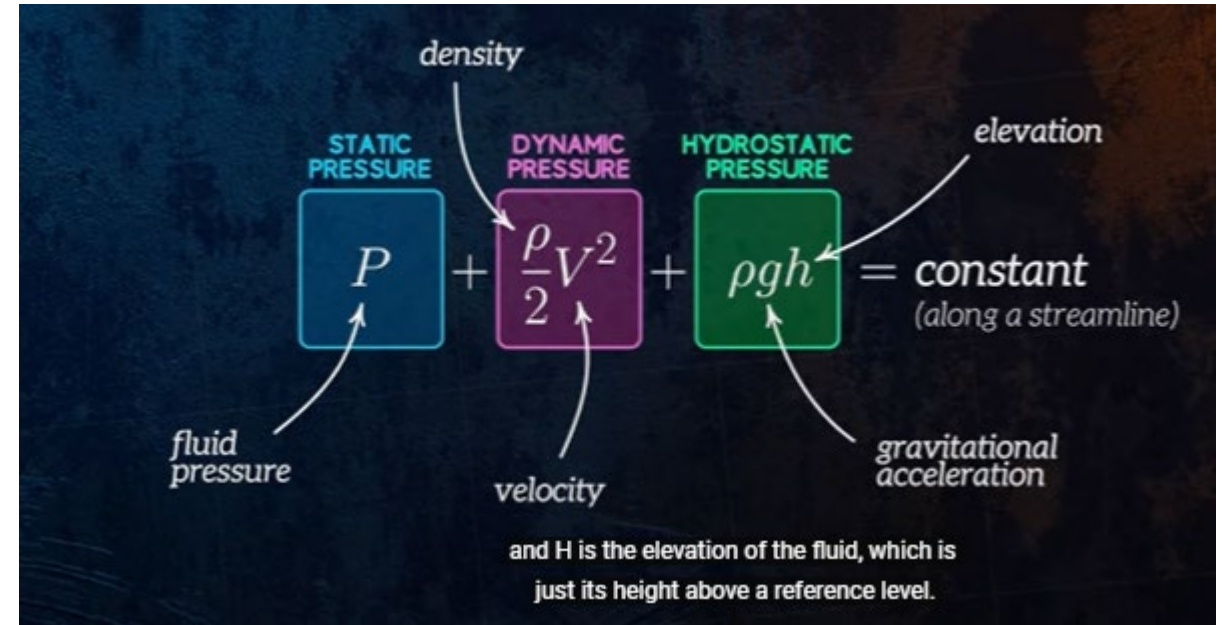
$$\begin{array}{c} \text{STATIC} \\ \text{PRESSURE} \end{array} \quad \begin{array}{c} \text{DYNAMIC} \\ \text{PRESSURE} \end{array} \quad \begin{array}{c} \text{HYDROSTATIC} \\ \text{PRESSURE} \end{array}$$

$$P + \frac{\rho}{2}V^2 + \rho gh = \text{constant} \quad (\text{along a streamline})$$

HEAD FORM

$$\frac{P}{\rho g} + \frac{V^2}{2g} + h = \text{constant} \quad (\text{along a streamline})$$

and the energy form.
We can think of Bernoulli's

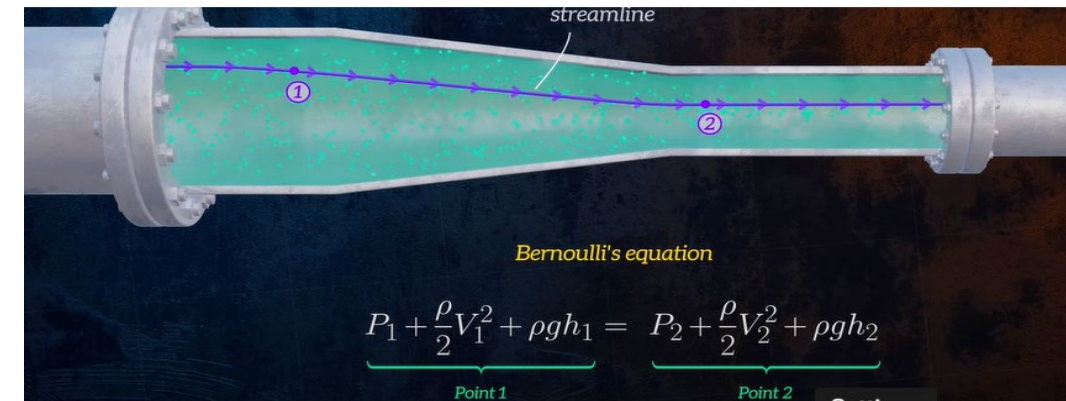


For a horizontal flow ($H_1 = H_2$)

There is no significant change in elevation between points 1 and 2 :

$$P_1 + \frac{\rho}{2}V_1^2 + \cancel{\rho gh_1} = P_2 + \frac{\rho}{2}V_2^2 + \cancel{\rho gh_2}$$

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$



If we assume the fluid is incompressible, then the mass flow rate at points 1 & 2 must be equal.

- This gives us what is called the **Continuity Equation**, which is just a statement of conservation of mass. Mass flow rate is equal to the product of the fluid density, the pipe cross-sectional area and the fluid velocity

Continuity equation
(conservation of mass)

$$\dot{m}_1 = \dot{m}_2$$

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

Continuity equation
(conservation of mass)

$$\rho A_1 V_1 = \rho A_2 V_2$$

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

we can rearrange the cont. Eq to find the velocity at the 2nd point

Continuity equation
(conservation of mass)

$$A_1 V_1 = A_2 V_2$$

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2}(V_2^2 - V_1^2)$$

$$V_2 = \frac{A_1}{A_2} V_1$$

The cross-sectional area at point 2 (A_2) is less than A_1 . which means the velocity of the flow increases as it passes into the smaller diameter pipe. This is quite intuitive.

$$V_2 > V_1 \quad V_2 = \frac{A_1}{A_2} V_1$$

Substituting V_2 into Bernoulli's Equation, we can see since the velocity increases between points 1 and 2, the pressure b/n both points must decrease.(pressure decrease b/n point 1 and 2)

Bernoulli's equation

$$P_1 - P_2 = \frac{\rho}{2} V_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right)$$

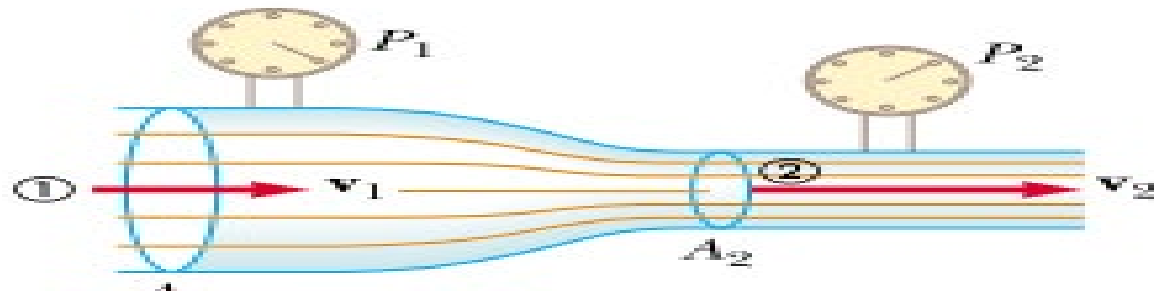
This concept, that for a horizontal flow an increase in fluid velocity must be accompanied by a decrease in pressure, is one way of formulating Bernoulli's principle.

BERNOULLI'S PRINCIPLE

For horizontal flow an **increase in velocity** must be accompanied by a **decrease in pressure**

Example (14.9)

1. In the horizontal constricted pipe illustrated in Fig below, determine the flow speed at point 2 if the pressure difference P_1 & P_2 is known.



Solution Because the pipe is horizontal, $y_1 = y_2$, and applying Equation 14.8 to points 1 and 2 gives

$$(1) \quad P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

From the equation of continuity, $A_1 v_1 = A_2 v_2$, we find that

$$(2) \quad v_1 = \frac{A_2}{A_1} v_2$$

Substituting this expression into Equation (1) gives

$$P_1 + \frac{1}{2}\rho \left(\frac{A_2}{A_1} \right)^2 v_2^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$v_2 = A_1 \sqrt{\frac{2(P_1 - P_2)}{\rho(A_1^2 - A_2^2)}}$$

Example (14.10)

2. An enclosed tank containing a liquid of density ρ has a hole in its side at a distance y_1 from the tank's bottom (Fig below). The hole is open to the atmosphere, and its diameter is much smaller than the diameter of the tank. The air above the liquid is maintained at a pressure P . Determine the speed of the liquid as it leaves the hole when the liquid's level is a distance h above the hole.

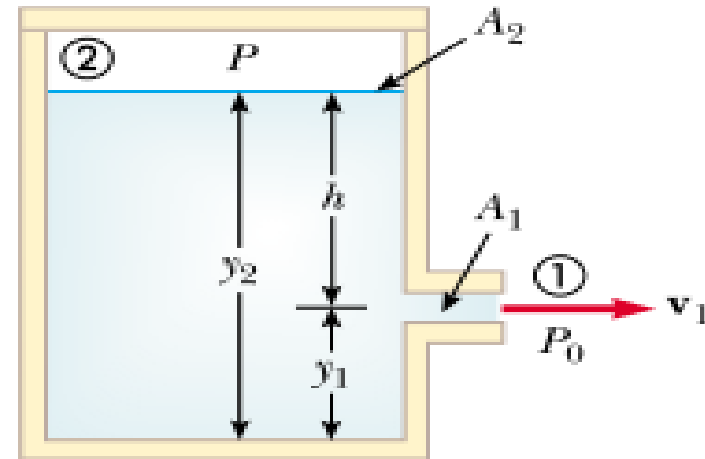
Solution

Because $A_2 \gg A_1$, (i.e $V_2 \ll V_1$), the liquid is approximately at rest at the top of the tank, where the pressure is P . Applying Bernoulli's equation to points 1 and 2 and noting that at the hole P_1 is equal to atmospheric pressure P_0 , we find that

$$P_0 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P + \rho g y_2$$

But $y_2 - y_1 = h$; thus, this expression reduces to

$$v_1 = \sqrt{\frac{2(P - P_0)}{\rho} + 2gh}$$



CHAPTER – 4

Heat & Thermodynamics



Addis Ababa Science & Technology University

Department of Physics

General Physics (phys1001)

Chapter - 4

Heat & Thermodynamics

July, 2022

Heat & Thermodynamics

Thermodynamics :

- is a branch of [physics](#) that deals with [heat](#), [work](#), and [temperature](#), and their relation to [energy](#), [entropy](#), and the physical properties of [matter](#) and [radiation](#).
- In broad terms, it deals with the transfer of energy from one system to another and from one form to another.

Thermodynamic system :

- is a definite quantity of gas enclosed in a cylinder with a sliding piston that allows the volume of gas to vary.
- In general, is a quantity of matter of fixed identity that is the subject of study and it may be solid, liquid or gas.

The surrounding :

- is the environment that is around a system and in thermal contact with it.
- In general, is defined by its temperature, volume, pressure, and chemical composition.

Note: A system is in equilibrium when each of these variables has the same value at all points.

A state of a thermodynamic system can be described by thermodynamic state variables.

A widely used distinction is between isolated, closed, and open thermodynamic systems.

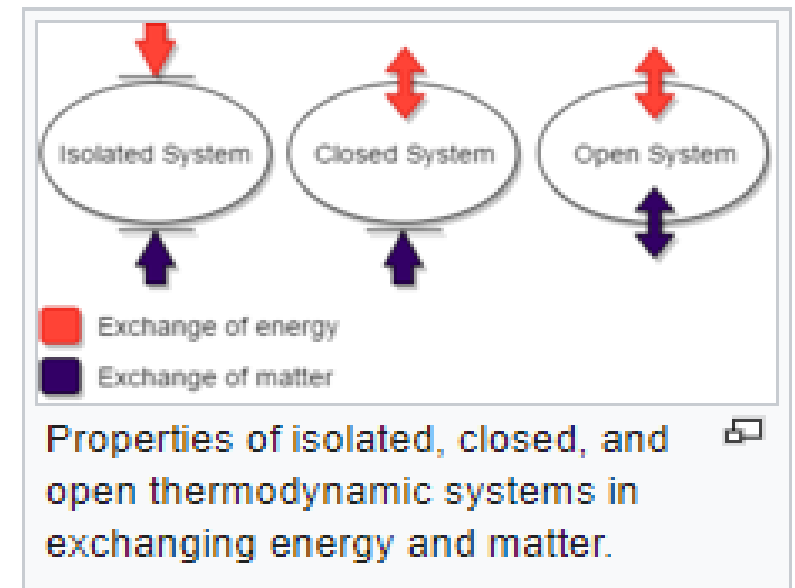
i) Isolated thermodynamic system:

- has walls that are non-conductive of heat and perfectly reflective of all radiation, that are rigid and immovable, and that are impermeable to all forms of matter and all forces. (Some writers use the word 'closed' when here the word 'isolated' is being used.)

ii) Closed thermodynamic system:

- is confined by walls that are impermeable to matter, but, by thermodynamic operations, alternately can be made permeable (described as 'diathermal') or impermeable ('adiabatic') to heat, and that, for thermodynamic processes (initiated and terminated by thermodynamic operations), alternately can be allowed or not allowed to move, with system volume change or agitation with internal friction in system contents, as in Joule's original demonstration of the mechanical equivalent of heat, and alternately can be made rough or smooth, so as to allow or not allow heating of the system by friction on its surface.

iii) Open thermodynamic system: has at least one wall that separates it from another thermodynamic system, which for this purpose is counted as part of the surroundings of the open system, the wall being permeable to at least one chemical substance, as well as to radiation; such a wall, when the open system is in thermodynamic equilibrium, does not sustain a temperature difference across itself.



Heat (Q):

- the transfer of energy across the boundary of a system due to a temperature difference between the system and its surroundings
- is defined as the flow of energy from one object to another.
- Flows from higher temperature to lower temperature
- *Heat leads to changes in internal energy and thus to changes in temperature, which cause the expansion or contraction of matter*
- *is a **spontaneous flow of energy** into or out of a system caused by a difference in temperature between the system and its surroundings, or between two objects whose temperatures are different*
- *the exchange of energy occurs due to the individual interactions of the particles.*
- *No macroscopic displacement occurs when heat flows and no macroscopic force is exerted by one object on the other.*
- *A system cannot possess heat or work; these two are energies that flow into or out of a system*
- *a body never contains heat. Rather, heat can be identified only as it crosses the boundary. Thus, heat is a transient phenomenon.*

N:B Heat transfer obeys the law of conservation of energy (if no heat is lost to the surroundings):

$$Q_{\text{lost by hotter object}} = Q_{\text{gained by cooler object}}$$

$$Q = Cm\Delta T$$

Work (W)

- is a **non-spontaneous transfer of energy** into or out of a system due to force acting through a displacement.
- it takes many forms, moving a piston, or stirring, or running an electrical current through a resistance.

Note: Heat and work are two possible ways of transferring energy from one system to another.

Temperature(T):

- is the measure of the average molecular kinetic energy of a substance
- determines the direction of heat flow (i.e., Heat flows from higher temperature to lower temperature)
- how hot or cold an object feels when we touch it (qualitative indication of temperature)
- determines whether or not energy will transfer between two objects in thermal contact.
- is an *intensive* quantity. Intensive quantities are independent of size.

Thermal equilibrium.

- Two objects are in thermal if they have the same temperature and no net energy is transferred between them.
- is a situation in which two objects would not exchange energy by heat or electromagnetic radiation when they are in thermal contact.

if $T_A = T_B$, then A and B are in thermal equilibrium

Internal Energy (E or U):

- is all the energy of a system that is associated with its **microscopic components—atoms and molecules**—when viewed from a reference frame at rest with respect to the **center of mass of the system**.
- The last part of this sentence ensures that any bulk kinetic energy of the system due to its motion through **space** is not included in internal energy.
- Internal energy includes : kinetic energy of random translational, rotational, and vibrational motion of molecules, potential energy within molecules, and potential energy between molecules.
- It is useful to relate internal energy to the temperature of an object, but this relationship is that internal energy changes can also occur in the absence of temperature changes.

Thermal Energy and Bond Energy

- **Thermal energy** : can be interpreted as that part of the internal energy associated with random motion of molecules and, therefore, related to temperature.
- **Bond energy** : is the intermolecular potential energy.

thus

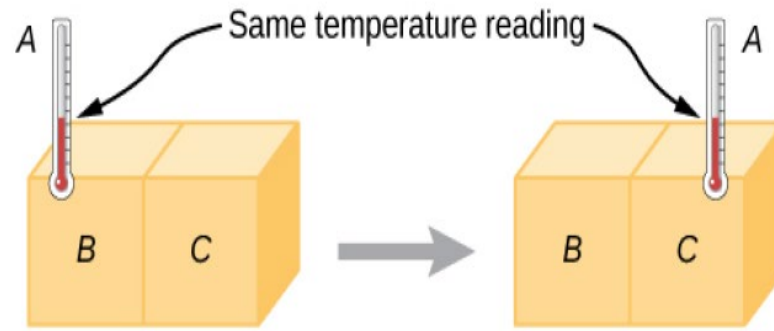
$$\text{internal energy} = \text{thermal energy} + \text{bond energy}$$

Laws of thermodynamics

1. Zeroth law of Thermodynamics

- states that “if object A is in thermal equilibrium with object B, and object B is in thermal equilibrium with object C, then object A is in thermal equilibrium with object C.”

i.e *If two systems are each in thermal equilibrium with a third, they are also in thermal equilibrium with each other.*



Temperature Scales

- **Thermometer:** measures temperature according to well-defined scales of measurement
- There are three most common temperature scales : **Fahrenheit, Celsius, and Kelvin.**
- **at STP:**
 - the boiling point of water = $100^{\circ}\text{C} = 212^{\circ}\text{F} = 373^{\circ}\text{K}$
 - the freezing point of water = $0^{\circ}\text{C} = 32^{\circ}\text{F} = 273^{\circ}\text{K}$

Relationships between Temperature Scales

To convert from...	Use this equation...
Celsius to Fahrenheit	$T_{\text{F}} = \frac{9}{5}T_{\text{C}} + 32$
Fahrenheit to Celsius	$T_{\text{C}} = \frac{5}{9}(T_{\text{F}} - 32)$
Celsius to Kelvin	$T_{\text{K}} = T_{\text{C}} + 273.15$
Kelvin to Celsius	$T_{\text{C}} = T_{\text{K}} - 273.15$
Fahrenheit to Kelvin	$T_{\text{K}} = \frac{5}{9}(T_{\text{F}} - 32) + 273.15$
Kelvin to Fahrenheit	$T_{\text{F}} = \frac{9}{5}(T_{\text{K}} - 273.15) + 32$

	$^{\circ}\text{F}$	$^{\circ}\text{C}$	K
Boiling point of water	212	100	373
Freezing point of water	32	0	273
Freezing point of dry ice (CO_2)	-109	-78	195
Boiling point of nitrogen	-321	-196	77
Absolute zero	-460	-273	0

Thermal Expansion

- is the change in size or volume of a given system as its temperature changes

1. Linear Thermal Expansion:

- The increase in length Δl of a solid is proportional to its initial length l_o and the change in its temperature ΔT .

$$\begin{aligned}\Delta l &\sim l_o \Delta T \\ \Delta l &= \alpha l_o \Delta T \quad \Rightarrow l_f - l_o = \alpha l_o \Delta T \quad \Rightarrow l_f = l_o + \alpha l_o \Delta T = l_o(1 + \alpha \Delta T) \\ \alpha &= \frac{\Delta l}{l_o \Delta T}, \quad \text{is called coefficient of linear expansion and its SI unit} = \frac{1}{K} = K^{-1} = (C^0)^{-1}\end{aligned}$$

Where., l_o = the initial length, T_o = initial temperature, l_f = final length
 $\Delta T = T - T_o$

Example:

1. A segment of steel railroad track has a length of 30.0 m when the temperature is 0.0°C . What is its length when the temperature is 40.0°C ? use : $\alpha = 11 \times 10^{-6} (^{\circ}\text{C})^{-1}$

2. Areal Expansion

The change in area ΔA of a solid is proportional to its initial area A_0 and the change in its temperature ΔT . That is,

$$\Delta A = \beta A_0 \Delta T \Rightarrow A = A_0(1 + \beta \Delta T) \text{ , where } \beta = 2\alpha \text{ is the coefficient of areal expansion.}$$

3. Volume Expansion

The change in volume ΔV of a solid is proportional to its initial volume V_0 and the change in its temperature ΔT . That is:

$$\Delta V = \gamma V_0 \Delta T \Rightarrow V = V_0(1 + \gamma \Delta T) \text{ , where } \gamma = 3\alpha \text{ is the coefficient of volume expansion.}$$

Specific Heat and Latent Heat

1. Specific Heats(Specific Heat Capacity) [**c** → small letter]:

- The quantity of heat, **Q**, required to change the temperature of a body of mass **m** by ΔT is proportional to both the mass, **m** and the change in temperature, ΔT . And it is given by:

$$Q \sim m\Delta T \Rightarrow Q = mc\Delta T$$

Where, $c = \frac{Q}{m\Delta T}$, is specific heat capacity which is defined as the amount of the heat required to raise the temperature of a 1kg mass of a substance by $1^\circ C$. Its SI unit = $\frac{J}{kg.K}$ or $\frac{J}{Kg. ^\circ C}$

2. Heat capacity(**C**) [**C** → Capital letter]:

- is defined as the amount of heat energy required to raise the temperature of a substance by $1^\circ C$

$$C = \frac{Q}{\Delta T} \Rightarrow Q = C\Delta T$$

Or

$$C = cM$$

- The amount of heat required to change the temperature of **n moles** of a substance, usually for gases, by ΔT is:

$$Q = nC\Delta T$$

Where, n = number of mole
 C = Heat capacity

Latent Heats (Hidden Heat)

- Is the heat required per unit mass of a substance to produce a phase change at constant temperature.
- The energy required to change the phase of a pure substance of mass m at a constant temperature is given by:

$$Q = \pm mL$$

where L is the latent heat of the substance and depends on the nature of the phase change and the properties of the substance. The positive sign is used if energy is entering the system, and the negative sign is used if energy is leaving.

Types of Latent Heat Transfer

There are two types of latent heat transfers between an object and its environment.

Latent Heat of Fusion (L_f): is the heat absorbed or released when matter melts, changing phase from solid to liquid form at constant temperature. For example, 333.7 kJ of heat is required to change 1 kg of ice to water at 0°C, so for water $L_f = 333.7 \text{ kJ/kg}$.

Latent Heat of Vaporization (L_v): is the heat absorbed or released when matter vaporizes, changing phase from liquid to gas phase at constant temperature. To change 1 kg of water to steam at 100°C, 2256 kJ of heat is required and so $L_v = 2256 \text{ kJ}$.

Example 1:

How much heat energy is required to change a 40 g ice cube from a solid at -10 °C to steam at 110 °C?

Solution:

To raise the temperature of the ice to °C we need,

$$\Delta Q_{ice} = m_{ice} c_{ice} \Delta T = 0.04 \text{ kg} \left(0.49 \frac{\text{kcal}}{\text{kg} \cdot ^\circ\text{C}} \right) 10 ^\circ\text{C} = 0.196 \text{ kcal}.$$

To melt the ice we need,

$$\Delta Q_{ice} = m_{ice} L_{ice} = 0.04 \text{ kg} (80 \text{ kcal/kg}) = 3.2 \text{ kcal}.$$

To raise the temperature of the water to 100 °C we need,

$$\Delta Q_{water} = m_{water} c_{water} \Delta T = 0.04 \text{ kg} \left(1 \frac{\text{kcal}}{\text{kg} \cdot ^\circ\text{C}} \right) 100 ^\circ\text{C} = 4 \text{ kcal}.$$

To boil the water we need,

$$\Delta Q_L = m_{water} L_{iwater} = 0.04 \text{ kg} \left(540 \text{ k} \frac{\text{cal}}{\text{kg}} \right) = 21.6 \text{ kcal}$$

To raise the temperature of the steam to 110°C we need,

$$\Delta Q_{ice} = m_{steam} c_{steam} \Delta T = 0.04 \text{ kg} (0.48 \text{ kcal}/(\text{kg}^\circ\text{C})) 10^\circ\text{C} = 0.192 \text{ kcal}.$$

Therefore, the total heat energy required is

$$\Delta Q = (0.196 + 3.2 + 4 + 21.6 + 0.192) \text{ kcal} = 29.188 \text{ kcal}.$$

Example 2. How much energy is required to convert a 1.00-g cube of ice at & 30.0°C to steam at 120.0°C ?

Part A. On this portion of the curve, the temperature of the ice changes from -30.0°C to 0.0°C . Because the specific heat of ice is $2\,090\text{ J/kg}\cdot^{\circ}\text{C}$, we can calculate the amount of energy added by using Equation 20.4:

$$Q = m_i c_i \Delta T = (1.00 \times 10^{-3} \text{ kg}) (2\,090 \text{ J/kg}\cdot^{\circ}\text{C}) (30.0^{\circ}\text{C}) = 62.7 \text{ J}$$

Part B. When the temperature of the ice reaches 0.0°C , the ice–water mixture remains at this temperature—even though energy is being added—until all the ice melts. The energy required to melt 1.00 g of ice at 0.0°C is, from Equation 20.6,

$$Q = m_i L_f = (1.00 \times 10^{-3} \text{ kg}) (3.33 \times 10^5 \text{ J/kg}) = 333 \text{ J}$$

Thus, we have moved to the 396 J ($= 62.7 \text{ J} + 333 \text{ J}$) mark on the energy axis in Figure 20.2.

Part C. Between 0.0°C and 100.0°C , nothing surprising happens. No phase change occurs, and so all energy added to the water is used to increase its temperature. The amount of energy necessary to increase the temperature from 0.0°C to 100.0°C is

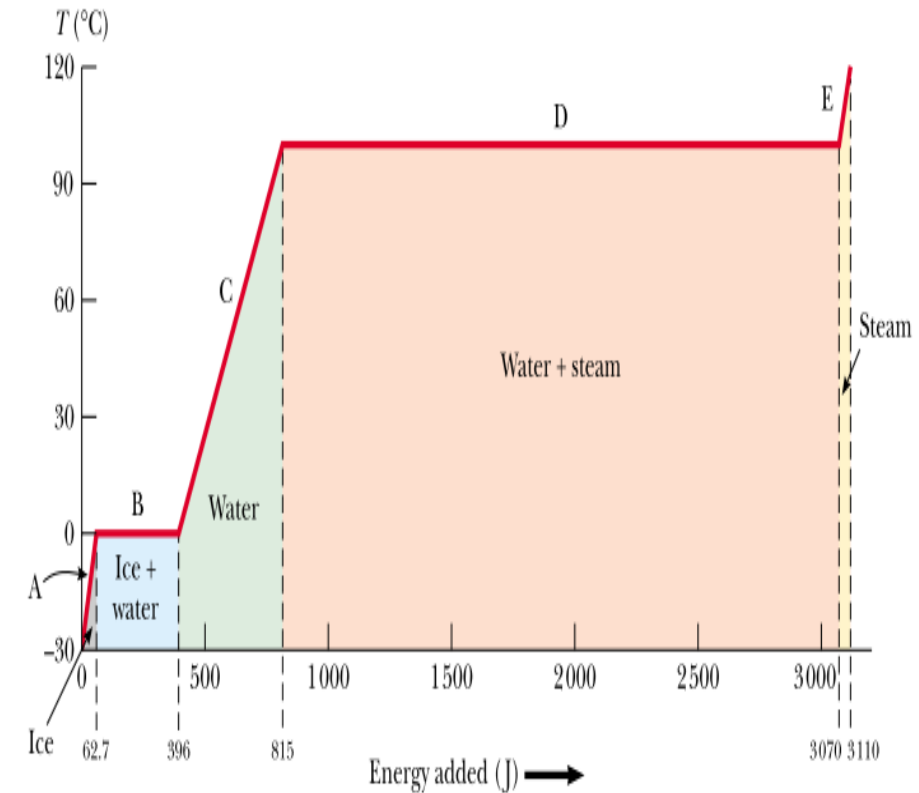
$$Q = m_w c_w \Delta T = (1.00 \times 10^{-3} \text{ kg}) (4.19 \times 10^3 \text{ J/kg}\cdot^{\circ}\text{C}) (100.0^{\circ}\text{C}) = 419 \text{ J}$$

Part D. At 100.0°C , another phase change occurs as the water changes from water at 100.0°C to steam at 100.0°C . Similar to the ice–water mixture in part B, the water–steam mixture remains at 100.0°C —even though energy is being added—until all of the liquid has been converted to steam. The energy required to convert 1.00 g of water to steam at 100.0°C is

$$Q = m_w L_v = (1.00 \times 10^{-3} \text{ kg}) (2.26 \times 10^6 \text{ J/kg}) = 2.26 \times 10^3 \text{ J}$$

Part E. On this portion of the curve, as in parts A and C, no phase change occurs; thus, all energy added is used to increase the temperature of the steam. The energy that must be added to raise the temperature of the steam from 100.0°C to 120.0°C is

$$Q = m_s c_s \Delta T = (1.00 \times 10^{-3} \text{ kg}) (2.01 \times 10^3 \text{ J/kg}\cdot^{\circ}\text{C}) (20.0^{\circ}\text{C}) = 40.2 \text{ J}$$



$$\text{Total Heat} = Q_1 + Q_2 + Q_3 + Q_4 + Q_5 = 3110 \text{ J}$$

N:B. To cool 1 g of steam at 120.0°C to ice at & -30.0°C , we must remove 3110 J of energy.

Example 3:

If 90 g of molten lead at 327.3°C is poured into a 300 g casting form made of iron and initially at 20.0°C , what is the final temperature of the system? Assume no energy is lost to the environment.

Solution:

The melting point of lead is 327.3°C . Assume the final temperature of the system is T ,

- the amount of energy released by the lead as it solidifies is $\Delta Q_L = m_{\text{lead}} L_{\text{lead}} = 0.09 \text{ kg}(2.45 \times 10^4 \text{ J/kg}) = 2205 \text{ J}$, and
- the amount of energy released as it cools is

$$\Delta Q = m_{\text{lead}} c_{\text{lead}} \Delta T = 0.09 (128)(327.3 - T) = (11.52)(327.3 - T)$$

This energy is absorbed by the iron. For the iron we therefore have

$$2205 + (11.52)(327.3 - T) = m_{\text{iron}} c_{\text{iron}} \Delta T = 0.3 (448)(T - 20)$$

$$5975.5 - (11.52)T = 134 T - 2688$$

$$8663.5 = (145.52)T$$

$$T = 59.5^{\circ}\text{C}$$

Heat Transfer Mechanisms

Heat may be transferred from one place to another in three ways:

- conduction
- convection
- radiation

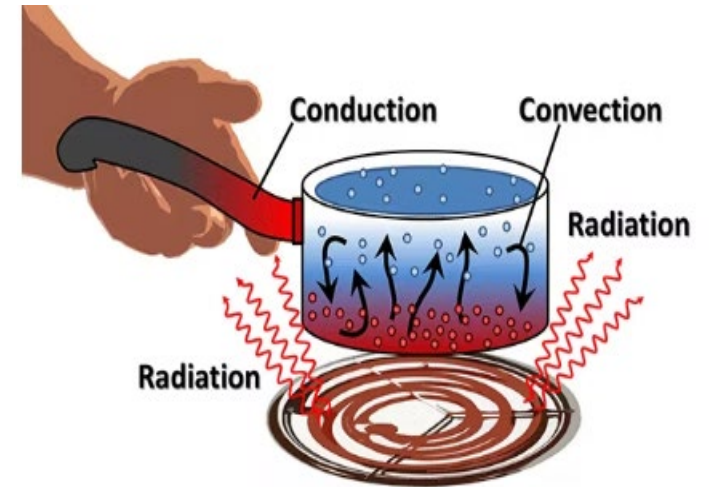
1. Conduction :

- transfers heat via direct molecular collision (occurs with physical contact)
- exchange of kinetic energy between colliding molecules or electrons
- is most obvious in solids.
- When a solid heats up, its particles gain kinetic energy and increase the energy with which they vibrate.
- occurs when heat energy travels through a body, passing from particle to particle as they vibrate & collide against each other.

For example,

- ✓ place your hand against a window or place metal into an open flame.

if you hold one end of a long metal bar and insert the other end into a flame, you will find that the temperature of the metal in your hand soon increases. The energy reaches your hand by means of conduction due to the collision of particles inside the metal.



2. Convection :

- Is the transfer of heat by the movement of the heated particles themselves.
- This can only take place in liquids and gases because in solids the particles are not able to move from their fixed positions.
- When a liquid or gas is heated, it expands and becomes less dense. The lighter liquid or gas rises allowing a flow of cooler material to take its place. This in turn becomes heated and so a current is set up. Heat will continue to be transferred through the available space in this way until it is evenly distributed.

Examples:

- ✓ Room is heated by a radiator.
 - The hot radiator warms the air in the lower regions of the room. The warm air expands and rises to the ceiling because of its lower density. The denser, cooler air from above sinks, and the continuous air current pattern shown in Fig
- ✓ Water heated in a teakettle
 - the lower layers are warmed first. This water expands and rises to the top because its density is lowered. At the same time, the denser, cool water at the surface sinks to the bottom of the kettle and is heated.

❖ *the warmer particle moved away from the source of heat to let the cooler ones collect the heat.*

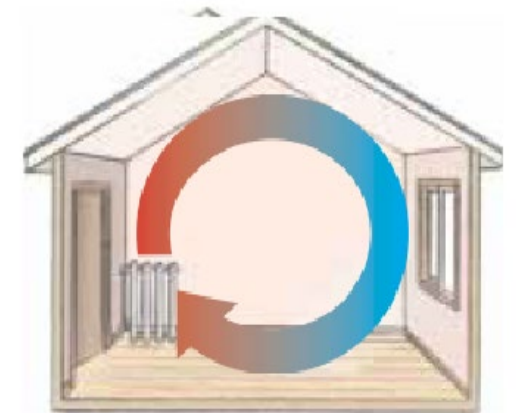
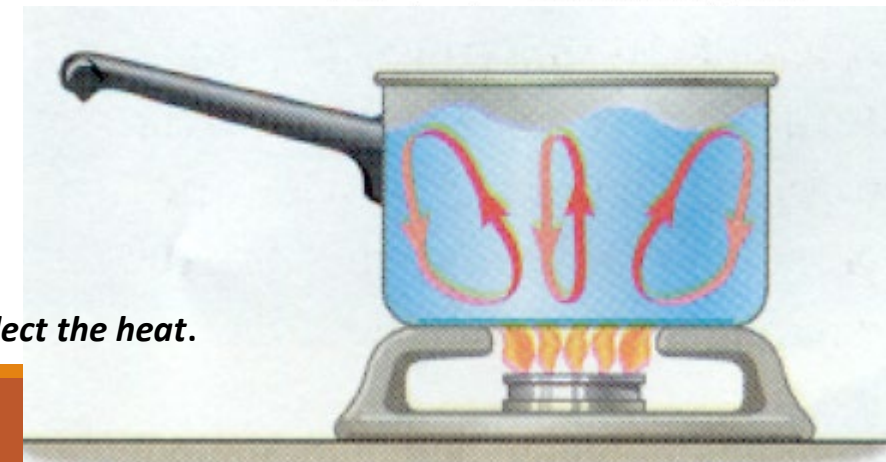
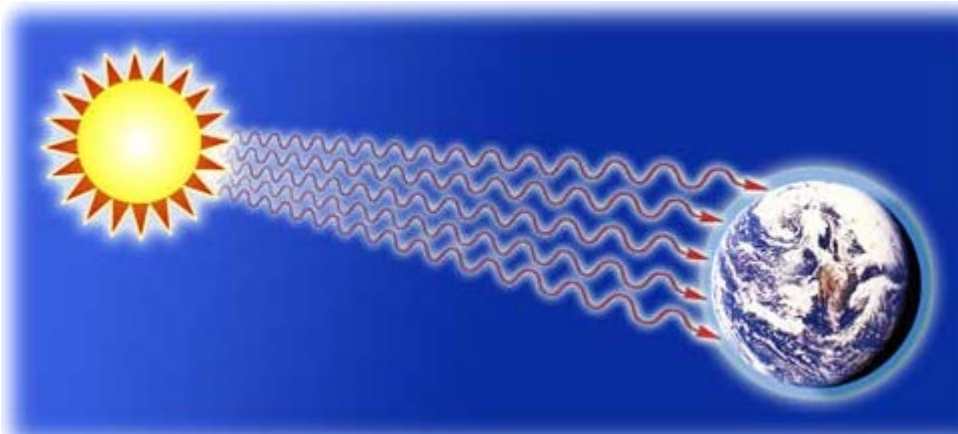


Fig 1 Convection currents

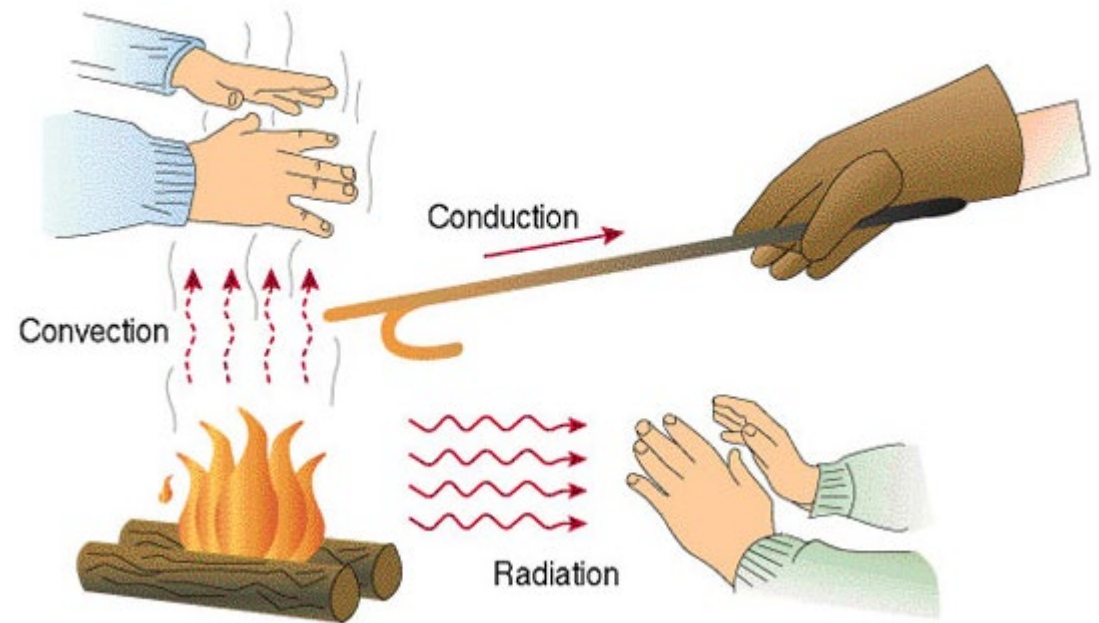


3. Radiation :

- Is the way we receive heat energy from the sun.
- It does not require a medium for its transmission (i.e. it can travel through empty space) and is in the form of electromagnetic energy waves which travel in the same way as light or radio waves



In reality, all 3 modes of heat transfer occur simultaneously.



Difference Between Conduction Convection and Radiation

Conduction	Convection	Radiation
In conduction, heat transfer occurs between objects by direct contact.	In convection, the heat transfer takes within the fluid.	In radiation, heat transfer occurs through electromagnetic waves without involving particles.
The heat transfer takes place due to the difference in temperature.	Heat transfer occurs due to the difference in density.	The heat transfer occurs in all objects with a temperature greater than 0 K.
Heat transfer in conduction is slow	Heat transfer in convection is faster.	Heat transfer in radiation is the fastest.
Heat transfer occurs through a heated solid object.	Heat transfer occurs through intermediate objects. For example, heat transfer between air and water.	Heat transfer occurs through electromagnetic waves.
It does not follow the law of reflection and refraction.	It does not follow the law of reflection and refraction.	It follows the law of reflection and refraction.

The First Law of Thermodynamics

- states that the change in internal energy of a system is equal to the sum of the heat flow into the system and the work done on the system.

$$\Delta U = Q + W$$

Where ΔU is the change in internal energy of the system, $W = P\Delta V$ is the work done on the system (or by the system) and P and ΔV are the pressure and change in volume of the system.

- The first law is a specialized statement of energy conservation applied to a thermodynamic system, such as a gas inside a cylinder that has a movable piston.

The First Law for different thermodynamic systems:

Isolated system-is a system which does not exchange heat with its surrounding and no work is done on the external environment. In this case $Q=0$ and $W=0$, so from the first law we conclude $\Delta U = 0$ or $U = \text{constant}$

The internal energy of an isolated system is constant.

Cyclic Process

Engines operate in cycles, in which the system –for example, a gas-periodically returns to its initial state. Since the system returns to its initial state, the change in internal energy in one complete cycle is zero; that is, $\Delta U = 0$. From the first law we see that

$$Q = W$$

- In a **cyclic process** (one that originates and terminates at the same state), $\Delta U = 0$ and, therefore, $Q + W = 0 \Rightarrow Q = -W$. That is, the energy transferred into the system by heat equals the negative of the work done on the system during the process.

Isochoric process

In a constant volume process, the volume of the system stays constant. Consequently, $W=0$.
From the first law we see that

$$\Delta U = Q$$

All the heat entering the system goes into increasing the internal energy.

Note: **Isovolumetric (Isochoric) Process** :

- is one that occurs at constant volume.
- No work is done in such a process ($W = 0$)

$$\text{so } \Delta U = Q$$

Adiabatic Process

In an adiabatic process, the system does not exchange heat with its surroundings; that is, $Q = 0$. The first law for an adiabatic process takes the form

$$\Delta U = W$$

- In an **adiabatic process**, no energy is transferred by heat between the system and its surroundings ($Q = 0$). In this case, the first law gives $\Delta U = W$. That is, the internal energy changes as a consequence of work being done on the system. In the adiabatic free expansion of a gas $Q = 0$ and $W = 0$, and so $\Delta U = 0$. That is, the internal energy of the gas does not change in such a process.

Isothermal Process

It is a process which involves no change in the temperature of the system. If the process occurs at constant temperature then there is no change in the internal energy of the system so $\Delta U = 0$. The first law for an isothermal process takes the form

$$\Delta U = Q + W$$

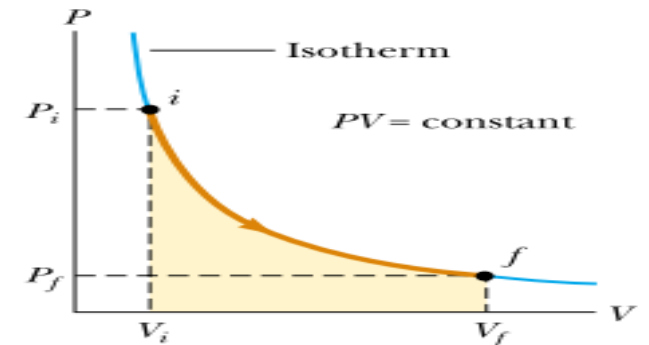
$$0 = Q + W$$

$$Q = -W$$

For an ideal gas in isothermal process the work done is calculated as

$$W = nRT \ln\left(\frac{V_i}{V_f}\right)$$

$$R = 8.31 \text{ J/mol.K} = \text{gas constant}$$



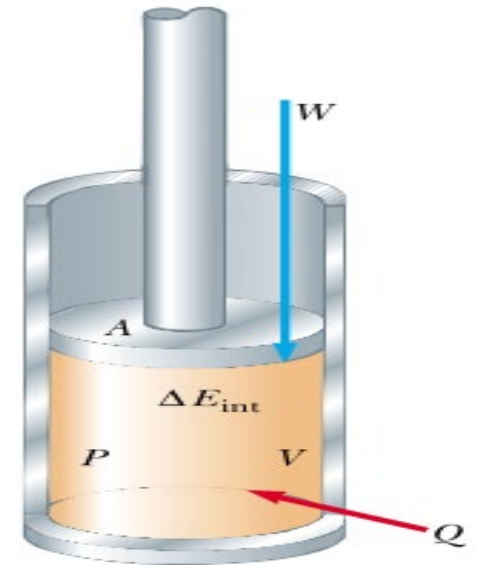
Isobaric process

In an isobaric process the expansion or compression occurs at constant pressure. Any work done by the system will result in an increase in volume. The work done in Pressure- Volume graph is equal to the area under the PV graph. For an isobaric process the work done W is calculated as

$$W = -P\Delta V = -P (V_f - V_i)$$

The first law for an isobaric process can be written as

$$\Delta U = Q + W \text{ or } \Delta U = Q - P\Delta V = Q - P (V_f - V_i)$$



Quick Quiz #1 Characterize the paths in Figure as isobaric, isovolumetric, isothermal, or adiabatic. Note that $Q = 0$ for path B.

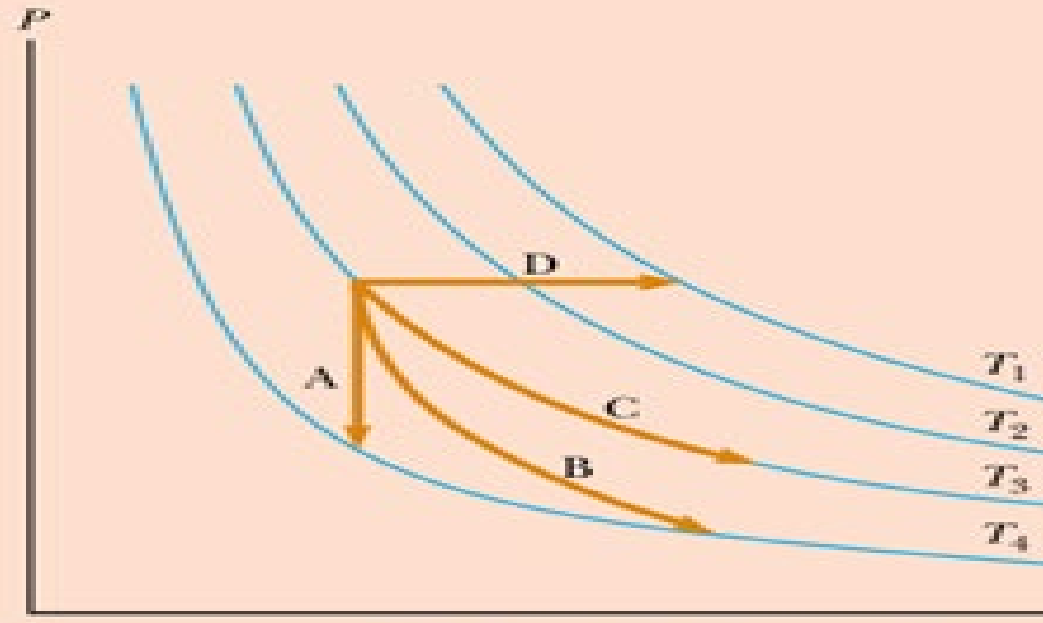


Figure Identify the nature of paths A, B, C, and D.

Answer:

A = Isovolumetric (Isochoric), D = Isobaric, C = Isothermal

B = Adiabatic, b/c $Q = 0$ implies, $\Delta U = W$ and From this result, we see that if a gas is compressed adiabatically such that W is positive, then ΔU is positive and the temperature of the gas increases. Conversely, the temperature of a gas decreases when the gas expands adiabatically.

Example 2:

2000 J of heat leaves the system and 3000 J of work is done on the system. What is the change in internal energy of the system?

Solution :

$$\Delta U = Q + W$$

$$\Delta U = -2000 + 3000$$

$$\Delta U = 1000 \text{ Joule}$$

Example3. A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L

A) How much work is done on the gas during the expansion?

B) How much energy transfer by heat occurs with the surroundings in this process?

C) If the gas is returned to the original volume by means of an isobaric process, how much work is done on the gas?

Solution

Given: $V_i = 3\text{ L}$, $V_f = 10\text{ L}$, $n = 1\text{ mol}$, $T_i = 0^\circ\text{C} = 273\text{ K} = \text{constant (Isothermal)}$, $R = \frac{8.31\text{ J}}{\text{mol}\cdot\text{K}}$, $1\text{ lit} = 1 \times 10^{-3}\text{ m}^3$

☐ **A.**
$$W = nRT \ln \left(\frac{V_i}{V_f} \right) = (1.0\text{ mol})(8.31\text{ J/mol}\cdot\text{K})(273\text{ K}) \ln \left(\frac{3.0\text{ L}}{10.0\text{ L}} \right)$$
$$= -2.7 \times 10^3\text{ J}$$

B. **Solution** From the first law, we find that

$$\Delta E_{\text{int}} = Q + W$$

$$0 = Q + W$$

$$Q = -W = 2.7 \times 10^3\text{ J}$$

$$\begin{aligned} W &= -P(V_f - V_i) = -\frac{nRT_i}{V_i} (V_f - V_i) \\ &= -\frac{(1.0\text{ mol})(8.31\text{ J/mol}\cdot\text{K})(273\text{ K})}{10.0 \times 10^{-3}\text{ m}^3} \\ &\quad \times (3.0 \times 10^{-3}\text{ m}^3 - 10.0 \times 10^{-3}\text{ m}^3) \\ &= 1.6 \times 10^3\text{ J} \end{aligned}$$

C. The work done in an isobaric process is given .

In this case, the initial volume is 10.0 L and the final volume is 3.0 L, the reverse of the situation in part (A). We are not given the pressure, so we need to incorporate the ideal gas law:

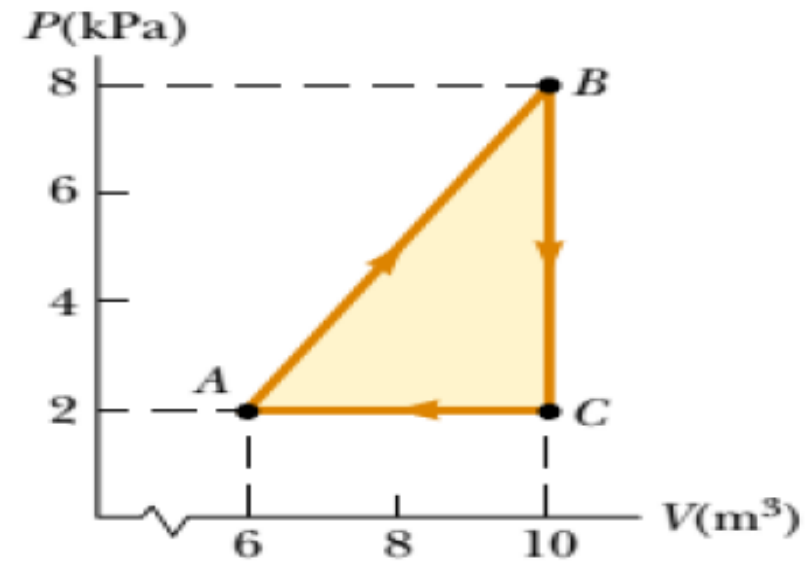
Notice that we use the initial temperature and volume to determine the value of the constant pressure because we do not know the final temperature. The work done on the gas is positive because the gas is being compressed.

Review Questions and Problems

1. Clearly distinguish among temperature, heat, and internal energy.
2. What is wrong with the following statement? “Given any two objects, the one with the higher temperature contains more heat.”
1. The temperature of a silver bar rises by 10°C when it absorbs 1.23kJ of energy by heat. The mass of the bar is 525g . Determine the specific heat of silver.
2. Calculate the quantity of heat required to raise the temperature of 1 g of ice from -10°C to 110°C .
3. If 20 g steam initially at 100°C is added to 60 g of ice initially at 0°C , then find the final equilibrium temperature of the mixture.
4. The temperature of a silver bar rises by 10.0°C when it absorbs 1.23 kJ of energy by heat. The mass of the bar is 525 g . Determine the specific heat of silver.
5. A 50.0-g sample of copper is at 25.0°C . If $1\ 200\text{ J}$ of energy is added to it by heat, what is the final temperature of the copper?

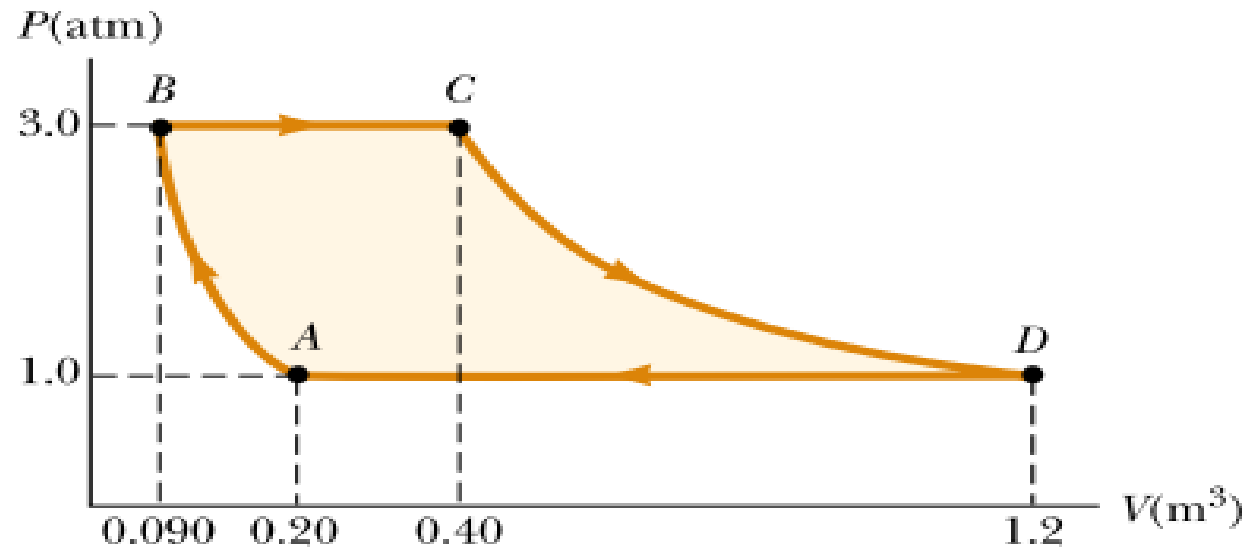
6. A system absorbs 35J of heat and in the process it does 11J of work. (a) If the initial internal energy is 205J what is the final internal energy? (b) The system follows a different thermodynamic path to the same final state and does 15J of work, what is the heat transferred?
7. A gas is compressed at a constant pressure of 0.800atm from 9.00 L to 2.00 L. In the process, 400 J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?
8. A gas is confined to a vertical cylinder by a piston of mass 2 kg and radius 1 cm. When 5J of heat are added, the piston rises by 2.4 cm. Find: (a) the work done by the gas; (b) the change in its internal energy. Atmospheric pressure is 10^5Pa .
9. A thermodynamic system undergoes a process in which its internal energy decreases by 500 J. At the same time, 220 J of work is done on the system. Find the energy transferred to or from it by heat.
10. A gas is compressed at a constant pressure of 0.8atm from 9.00L to 2.0 L. In the process, 400J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?

11. A gas is taken through the cyclic process described in Figure 1. (a) Find the net energy transferred to the system by heat during one complete cycle. (b) If the cycle is reversed-that is, the process follows the path ACBA-what is the net energy input per cycle by heat?

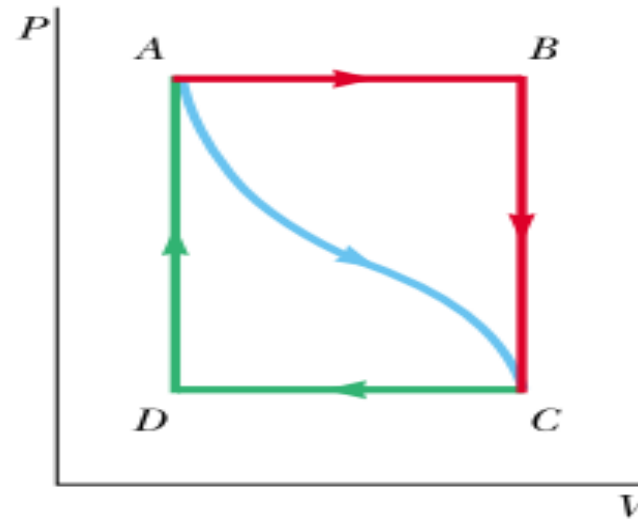


12. A sample of an ideal gas goes through the process shown in the Fig below. From A to B, the process is adiabatic; from B to C, it is isobaric with 100 kJ of energy entering the system by heat. From C to D, the process is isothermal; from D to A, it is isobaric with 150 kJ of energy leaving the system by heat. Determine the difference in internal energy

$$E_{int,B} - E_{int,A}$$

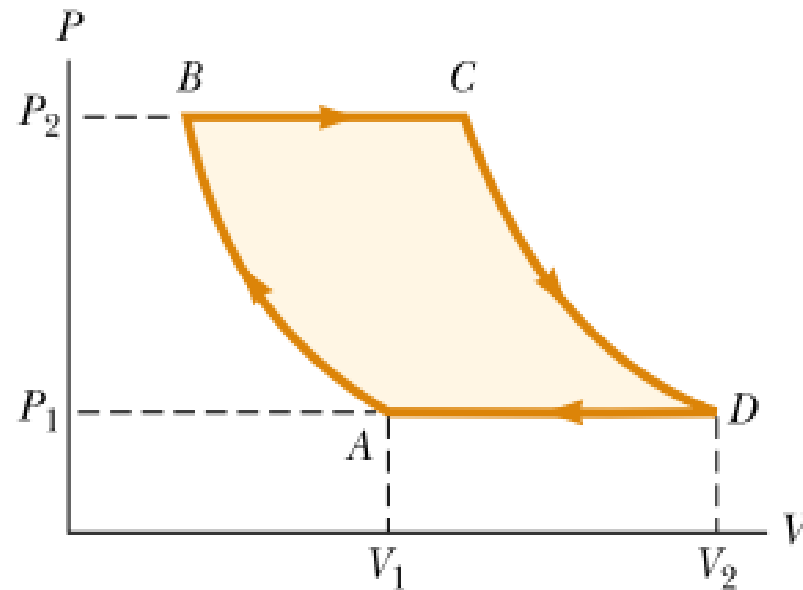


12. In the Fig. below, the change in internal energy of a gas that is taken from A to C is $+800\text{ J}$. The work done on the gas along path ABC is -500 J .
- How much energy must be added to the system by heat as it goes from A through B to C?
 - If the pressure at point A is five times that of point C, what is the work done on the system in going from C to D?
 - What is the energy exchanged with the surroundings by heat as the cycle goes from C to A along the green path?
 - If the change in internal energy in going from point D to point A is $+500\text{ J}$, how much energy must be added to the system by heat as it goes from point C to point D?



13. An ideal gas is carried through a thermodynamic cycle consisting of two isobaric and two isothermal processes as shown in Fig below. Show that the net work done on the gas in the entire cycle is given by

$$W_{\text{net}} = -P_1(V_2 - V_1) \ln \frac{P_2}{P_1}$$



Thank You !



Addis Ababa Science & Technology University

Department of Physics

General Physics (phys1001)

Chapter - 6

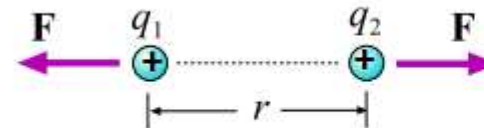
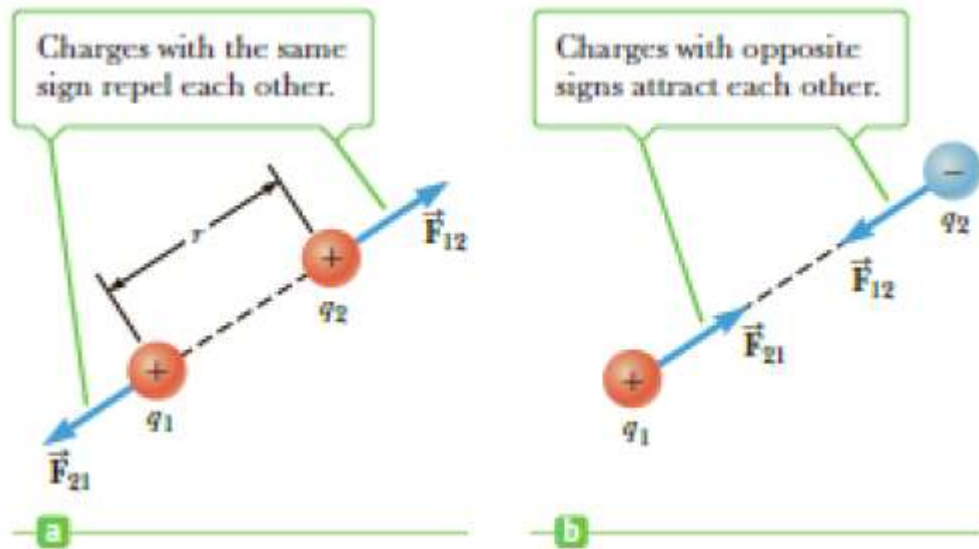
Electromagnetism

July, 2022

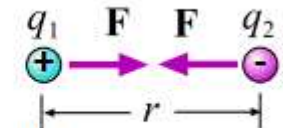
Electromagnetism

Electric Charge (Q or q):

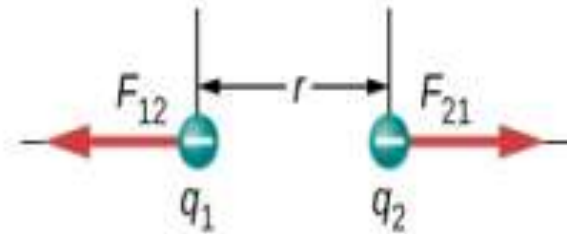
- is an inherent property of matter that makes it to have and experience electrical and magnetic characteristics.
- is two types:
 - negative charge
 - positive charge
- SI unit of charge is coulomb (C),
- The interaction of electric charges is governed by the following basic law of electrostatics which states that:
 - Like charges repel each other.
 - Unlike charges attract each other



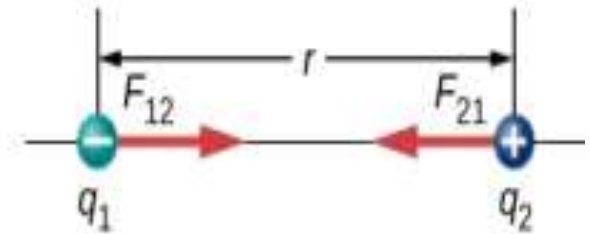
(a) same charge = repel each other



(b) opposite charge = attract each other



(a)



(b)

Coulomb's Law:

- States that If two point charges q_1 and q_2 are separated by a distance r , then the magnitude of the force of repulsion or attraction between them is directly proportional to the product of the magnitude of the charges and inversely proportional to the square the separation distance between them.

$$F = k \frac{|q_1| |q_2|}{r^2} \quad \text{where} \quad k = 8.9876 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2} \quad \text{OR} \quad k = \frac{1}{4\pi\epsilon_0} \quad \text{where} \quad \epsilon_0 = 8.85419 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$$

where K called the Coulomb constant. ϵ_0 is called the permittivity constant

Superposition Principle (Electric Forces due to system of point charges)

- Considering a system of N point charges, the net electric force on the test charge is simply the vector sum of each individual electric forces exerted on it by each of the individual point charges.

$$\vec{F} = k \sum_{i=1}^N \frac{q_i q_0}{r_{i0}^2} \hat{r}_{i0}$$

Example-1. The electron and proton of a hydrogen atom are separated by a distance of approximately $5.3 \times 10^{-11} \text{ m}$. Find the magnitudes of the electric force and the gravitational force between the two particles.

(Given: $e^- = -1.6 \times 10^{-19}$, $e^+ = +1.6 \times 10^{-19}$, $K = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$, $m_{e^-} = 9.1 \times 10^{-31} \text{ kg}$, $m_{e^+} = 1.6 \times 10^{-27} \text{ kg}$)

Solution From Coulomb's law, we find that the magnitude of the electric force is

$$F_e = k_e \frac{|e||-e|}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(1.60 \times 10^{-19} \text{ C})^2}{(5.3 \times 10^{-11} \text{ m})^2}$$
$$= 8.2 \times 10^{-8} \text{ N}$$

✓ The magnitude of gravitational force is:

$$F_g = G \frac{m_e m_p}{r^2}$$
$$= (6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)$$
$$\times \frac{(9.11 \times 10^{-31} \text{ kg})(1.67 \times 10^{-27} \text{ kg})}{(5.3 \times 10^{-11} \text{ m})^2}$$
$$= 3.6 \times 10^{-47} \text{ N}$$

Example-2

1. A charged particle A exerts a force of 2.62 N to the right on charged particle B when the particles are 13.7 mm apart. If the separation distance between them increases to 17.7 mm, what vector force does particle B then exert on A?

Solution:

By Colum's law the force the particles A and B exerted on each other is given by

$$F = \frac{k_e |q_B| |q_A|}{r^2}$$

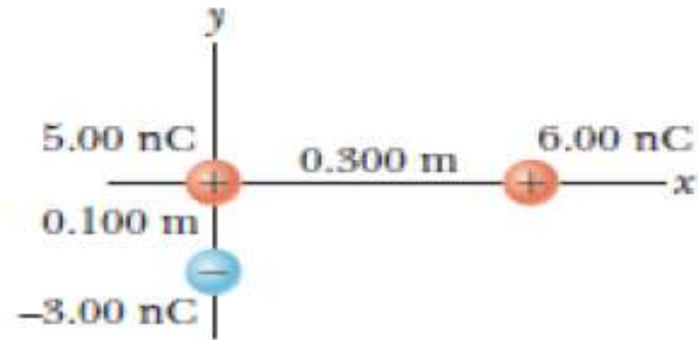
Hence, we can determine the product of the two charges as

$$|q_B| |q_A| = \frac{Fr^2}{k_e} = \frac{2.62 \text{ N} \times (1.37 \times 10^{-2} \text{ m})^2}{9.0 \times 10^9 \text{ Nm}^2 \text{C}^{-2}} = 5.44 \times 10^{-14} \text{ C}^2$$

Now we can use this result to determine the force in new position.

$$F = \frac{k_e |q_B| |q_A|}{r^2} = \frac{9.0 \times 10^9 \text{ Nm}^2 \text{C}^{-2} \times 5.44 \times 10^{-14} \text{ C}^2}{(1.77 \times 10^{-2} \text{ m})^2} = 1.56 \text{ N}$$

Example-3. Three point charges are arranged as shown in Fig. below Find
(a) the magnitude and
(b) the direction of the electric force on the particle at the origin.



Example-4

Consider three point charges located at the corners of a right triangle as shown in Figure 23.8, where $q_1 = q_3 = 5.0 \mu\text{C}$,

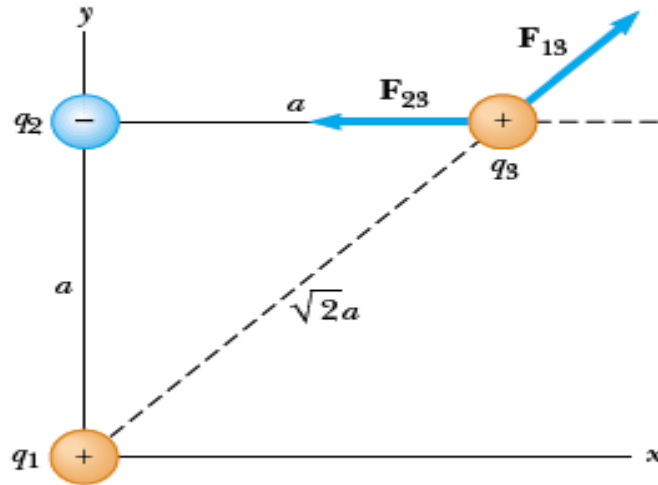


Figure 23.8 (Example 23.2) The force exerted by q_1 on q_3 is $\mathbf{F}_{13}$. The force exerted by q_2 on q_3 is $\mathbf{F}_{23}$. The resultant force $\mathbf{F}_3$ exerted on q_3 is the vector sum $\mathbf{F}_{13} + \mathbf{F}_{23}$.

$q_2 = -2.0 \mu\text{C}$, and $a = 0.10 \text{ m}$. Find the resultant force exerted on q_3 .

Solution First, note the direction of the individual forces exerted by q_1 and q_2 on q_3 . The force $\mathbf{F}_{23}$ exerted by q_2 on q_3 is attractive because q_2 and q_3 have opposite signs. The force $\mathbf{F}_{13}$ exerted by q_1 on q_3 is repulsive because both charges are positive.

The magnitude of $\mathbf{F}_{23}$ is

$$\begin{aligned} F_{23} &= k_e \frac{|q_2||q_3|}{a^2} \\ &= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(2.0 \times 10^{-6} \text{ C})(5.0 \times 10^{-6} \text{ C})}{(0.10 \text{ m})^2} \\ &= 9.0 \text{ N} \end{aligned}$$

In the coordinate system shown in Figure 23.8, the attractive force $\mathbf{F}_{23}$ is to the left (in the negative x direction).

The magnitude of the force $\mathbf{F}_{13}$ exerted by q_1 on q_3 is

$$\begin{aligned} F_{13} &= k_e \frac{|q_1||q_3|}{(\sqrt{2}a)^2} \\ &= (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \frac{(5.0 \times 10^{-6} \text{ C})(5.0 \times 10^{-6} \text{ C})}{2(0.10 \text{ m})^2} \\ &= 11 \text{ N} \end{aligned}$$

The repulsive force $\mathbf{F}_{13}$ makes an angle of 45° with the x axis. Therefore, the x and y components of $\mathbf{F}_{13}$ are equal, with magnitude given by $F_{13} \cos 45^\circ = 7.9 \text{ N}$.

Combining $\mathbf{F}_{13}$ with $\mathbf{F}_{23}$ by the rules of vector addition, we arrive at the x and y components of the resultant force acting on q_3 :

$$F_{3x} = F_{13x} + F_{23x} = 7.9 \text{ N} + (-9.0 \text{ N}) = -1.1 \text{ N}$$

$$F_{3y} = F_{13y} + F_{23y} = 7.9 \text{ N} + 0 = 7.9 \text{ N}$$

We can also express the resultant force acting on q_3 in unit-vector form as

$$\mathbf{F}_3 = (-1.1\hat{\mathbf{i}} + 7.9\hat{\mathbf{j}}) \text{ N}$$

What If? What if the signs of all three charges were changed to the opposite signs? How would this affect the result for $\mathbf{F}_3$?

Answer The charge q_3 would still be attracted toward q_2 and repelled from q_1 with forces of the same magnitude. Thus, the final result for $\mathbf{F}_3$ would be exactly the same.

Example-5. Three point charges lie along the x axis as shown in Fig. below

The positive charge $q_1 = 15.0 \mu\text{C}$ is at $x = 2.00 \text{ m}$, the positive charge $q_2 = 6.00 \mu\text{C}$ is at the origin, and the resultant force acting on q_3 is zero. What is the x coordinate of q_3 ?

Solution Because q_3 is negative and q_1 and q_2 are positive, the forces $\mathbf{F}_{13}$ and $\mathbf{F}_{23}$ are both attractive, as indicated in Figure 23.9. From Coulomb's law, $\mathbf{F}_{13}$ and $\mathbf{F}_{23}$ have magnitudes

$$F_{13} = k_e \frac{|q_1||q_3|}{(2.00 - x)^2} \quad F_{23} = k_e \frac{|q_2||q_3|}{x^2}$$

For the resultant force on q_3 to be zero, $\mathbf{F}_{23}$ must be equal in magnitude and opposite in direction to $\mathbf{F}_{13}$. Setting the magnitudes of the two forces equal, we have

$$k_e \frac{|q_2||q_3|}{x^2} = k_e \frac{|q_1||q_3|}{(2.00 - x)^2}$$

Noting that k_e and $|q_3|$ are common to both sides and so can be dropped, we solve for x and find that

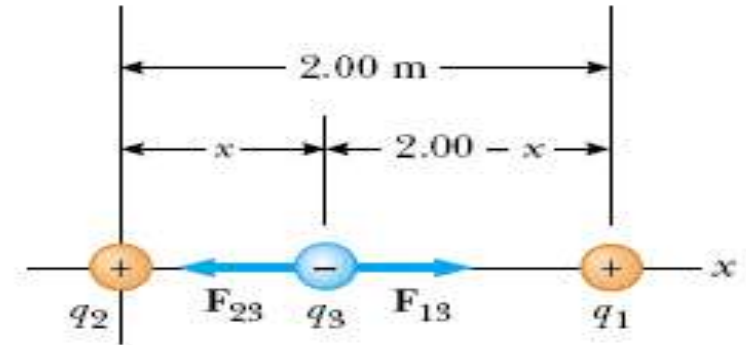
$$(2.00 - x)^2 |q_2| = x^2 |q_1|$$

$$(4.00 - 4.00x + x^2)(6.00 \times 10^{-6} \text{ C}) = x^2(15.0 \times 10^{-6} \text{ C})$$

This can be reduced to the following quadratic equation:

$$3.00x^2 + 8.00x - 8.00 = 0$$

Solving this quadratic equation for x , we find that the positive root is $x = 0.775 \text{ m}$. There is also a second root, $x = -3.44 \text{ m}$. This is another location at which the magnitudes

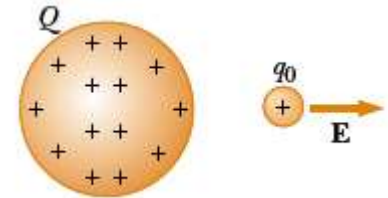


N:B. There is also a second root, $x = 3.44 \text{ m}$.
This is another location at which the magnitudes

The Electric Field

the electric field vector $\mathbf{E}$ at a point in space is defined as the electric force $\mathbf{F}_e$ acting on a positive test charge q_0 placed at that point divided by the test charge:

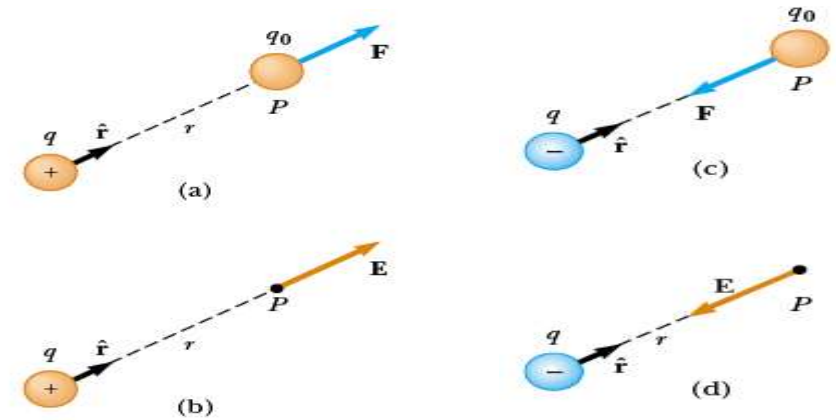
$$\mathbf{E} \equiv \frac{\mathbf{F}_e}{q_0}$$



$$\mathbf{F}_e = q\mathbf{E}$$

- The electric field created by a charge q at a point P at a distance r is given by:

$$\mathbf{E} = k_e \frac{q}{r^2} \hat{\mathbf{r}}$$



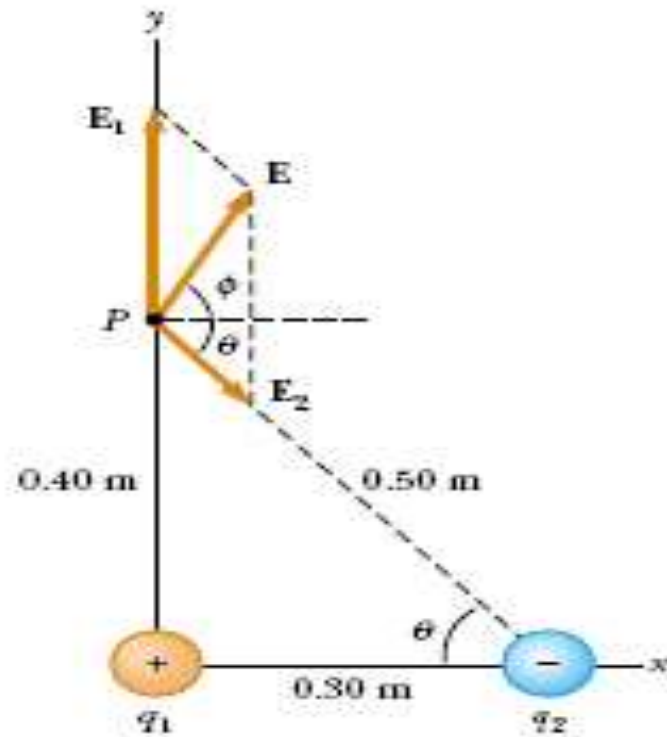
- The electric field at point P due to a group of source charges is given by :

$$\mathbf{E} = k_e \sum_i \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$

where r_i is the distance from the i th source charge q_i to the point P and $\hat{\mathbf{r}}_i$ is a unit vector directed from q_i toward P .

Example -1. A charge $q_1 = 7.0\mu\text{C}$ is located at the origin, and a second charge $q_2 = -5.0\mu\text{C}$ is located on the X -axis, 0.30 m from the origin (Fig.). Find the electric field at the point P , which has coordinates (0, 0.40) m.

Solution:



Solution First, let us find the magnitude of the electric field at P due to each charge. The fields E_1 due to the $7.0\mu\text{C}$ charge and E_2 due to the $-5.0\mu\text{C}$ charge are shown in Figure 23.14. Their magnitudes are

$$E_1 = k_e \frac{|q_1|}{r_1^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(7.0 \times 10^{-6} \text{ C})}{(0.40 \text{ m})^2} = 3.9 \times 10^5 \text{ N/C}$$

$$E_2 = k_e \frac{|q_2|}{r_2^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(5.0 \times 10^{-6} \text{ C})}{(0.50 \text{ m})^2} = 1.8 \times 10^5 \text{ N/C}$$

The vector E_1 has only a y component. The vector E_2 has an x component given by $E_2 \cos \theta = \frac{3}{5} E_2$ and a negative y component given by $-E_2 \sin \theta = -\frac{4}{5} E_2$. Hence, we can express the vectors as

$$E_1 = 3.9 \times 10^5 \hat{j} \text{ N/C}$$

$$E_2 = (1.1 \times 10^5 \hat{i} - 1.4 \times 10^5 \hat{j}) \text{ N/C}$$

The resultant field E at P is the superposition of E_1 and E_2 :

$$E = E_1 + E_2 = (1.1 \times 10^5 \hat{i} + 2.5 \times 10^5 \hat{j}) \text{ N/C}$$

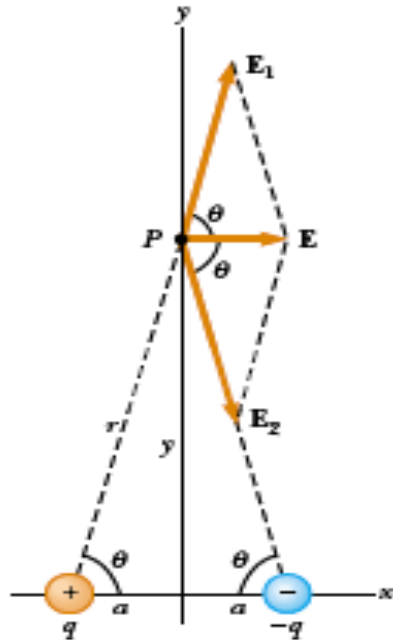
From this result, we find that E makes an angle ϕ of 66° with the positive x axis and has a magnitude of $2.7 \times 10^5 \text{ N/C}$.

Example-2. Electric Field of a Dipole

An electric dipole is defined as a positive charge q and a negative charge $-q$ separated by a distance $2a$. For the dipole shown in Fig. below, find the electric field $\mathbf{E}$ at P due to the dipole, where P is a distance $y \gg a$ from the origin.

Solution At P , the fields $\mathbf{E}_1$ and $\mathbf{E}_2$ due to the two charges are equal in magnitude because P is equidistant from the charges. The total field is $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$, where

$$E_1 = E_2 = k_e \frac{q}{r^2} = k_e \frac{q}{y^2 + a^2}$$



The y components of $\mathbf{E}_1$ and $\mathbf{E}_2$ cancel each other, and the x components are both in the positive x direction and have the same magnitude. Therefore, $\mathbf{E}$ is parallel to the x axis and has a magnitude equal to $2E_1 \cos \theta$. From Figure 23.15 we see that $\cos \theta = a/r = a/(y^2 + a^2)^{1/2}$. Therefore,

$$\begin{aligned} E &= 2E_1 \cos \theta = 2k_e \frac{q}{(y^2 + a^2)} \frac{a}{(y^2 + a^2)^{1/2}} \\ &= k_e \frac{2qa}{(y^2 + a^2)^{3/2}} \end{aligned}$$

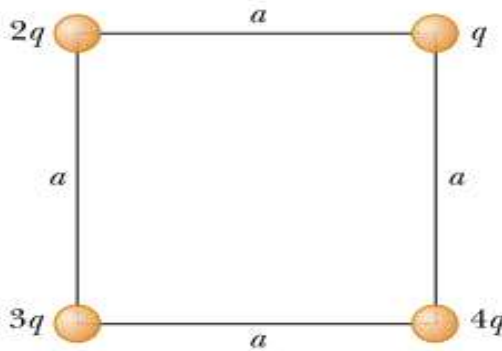
Because $y \gg a$, we can neglect a^2 compared to y^2 and write

$$E \approx k_e \frac{2qa}{y^3}$$

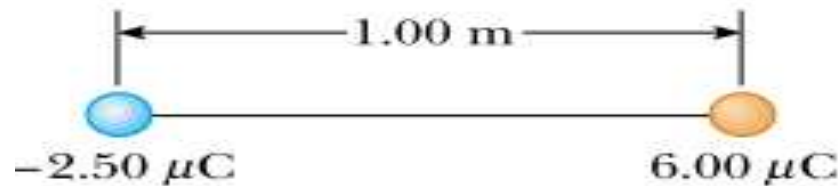
Exercise

1. Four point charges are at the corners of a square of side a as shown in Fig .below.

- (a) Determine the magnitude and direction of the electric field at the location of charge q .
- (b) What is the resultant force on q ?

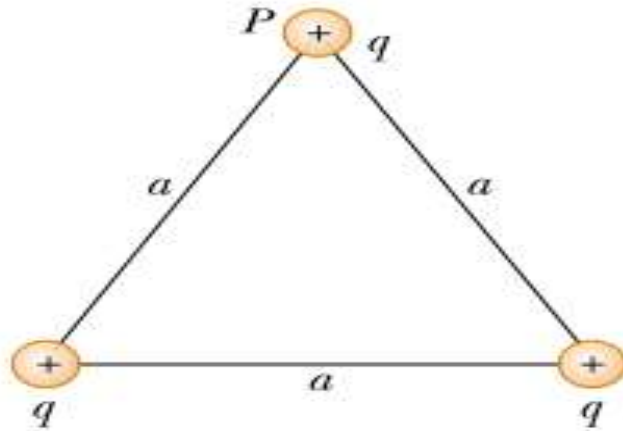


2. In Figure below , determine the point (other than infinity) at which the electric field is zero.



3. Three equal positive charges q are at the corners of an equilateral triangle of side a as shown in Fig. below. Find

- a) the location of a point (other than ∞) where the electric field is zero.
- b) What are the magnitude and direction of the electric field at P due to the two charges at the base?



Electric Potential Energy (U)

- A charge accelerated by an electric field is analogous to a mass going down a hill. In both cases, potential energy decreases as kinetic energy increases, $-\Delta U = \Delta K$. Work is done by a force, but since this force is conservative, we can write, $W = -\Delta U$
- Therefore, we define the electrical potential energy of any two systems of charges q_1 and q_2 is separated by a distance r_{12} is given by:

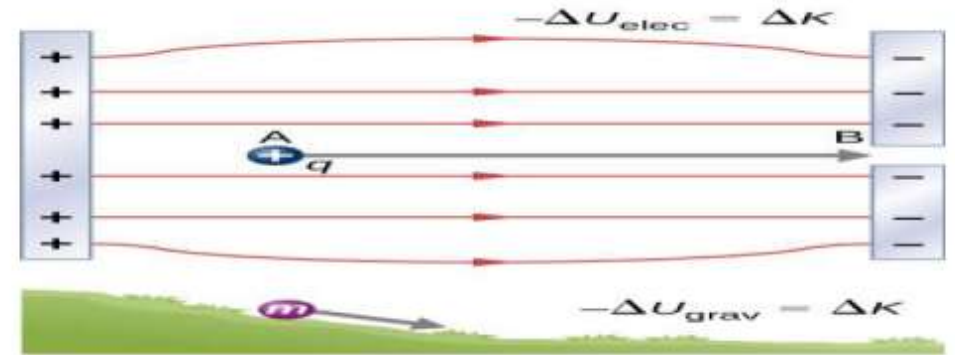
$$U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

- So the quantity U_{12} is the amount of work done on the two charges to bring them from infinity to the separation r_{12}
- For systems of charges, more than two, $q_1, q_2, q_3, \dots, q_n$, the electric potential energy is given by:

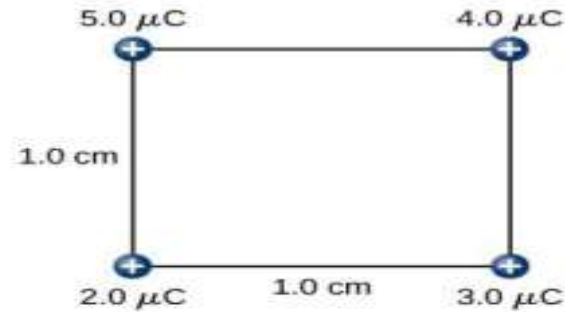
$$U = \sum_{i,j} \frac{1}{4\pi\epsilon_0} \left(\frac{q_i q_j}{r_{ij}} \right)$$

(superposition principle): the potential energy of a system of multiple charges is the sum of the potential energies of the individual pairs.

N:B. The sign of the charges will not be ignored if the charges are negative.



Example-1. Find the electric potential energy in assembling four charges at the vertices of a square of side 1.0 cm, starting each charge from infinity as shown in the figure



Solution:

From the formula of electric potential energy of systems of charges above,

$$\begin{aligned}
 U &= \sum_{i,j}^n \frac{1}{4\pi\epsilon_0} \left(\frac{q_i q_j}{r_{ij}} \right) \\
 &= \frac{1}{4\pi\epsilon_0} \left[\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \frac{q_3 q_4}{r_{34}} \right) \right] \\
 &= \\
 &= \frac{1}{4\pi \times 8.85 \times 10^{-12}} \left[\left(\frac{5 \times 2}{1 \times 10^{-2}} + \frac{5 \times 4}{1 \times 10^{-2}} + \frac{5 \times 3}{\sqrt{(1 \times 10^{-2})(1 \times 10^{-2})}} + \frac{4 \times 3}{1 \times 10^{-2}} + \frac{4 \times 2}{\sqrt{(1 \times 10^{-2})(1 \times 10^{-2})}} + \frac{3 \times 2}{1 \times 10^{-2}} \right) \times 10^{-9} \right]
 \end{aligned}$$

$$U = 57.8 \text{ J}$$

Electric Potential (V)

- Is the electric potential energy(U) per unit charge (q).
- Is potential difference V between two points A and B. i.e.

$$V = \frac{U}{q}$$

$$V = V_B - V_A = \frac{U}{q}$$

- SI Unit of potential difference (Voltage) is Joules per Coulomb called Volt (V)

$$1V = \frac{1J}{C}$$

Example -1. You have a 12.0-V motorcycle battery that can move 5000 C of charge, and a 12.0-V car battery that can move 60,000 C of charge. How much energy does each deliver?

Solution. $V = \frac{U}{q} \Rightarrow U = qV \quad \Rightarrow \quad U_1 = q_1V = (12V \times 5000C) = 60000J$

$$U_2 = q_2V = (12V \times 60,000C) = 720,000J = 0.7MJ$$

Current, Resistance and Ohm's Law

Current(I):

- is the rate of flow of charges in a given cross-sectional area.

$$I = \frac{\Delta Q}{\Delta t}$$

Where ΔQ is the amount of charge passing through a given area in time Δt .

- The SI unit for current is the ampere (A), and SI unit of charge is Coulomb (C)

$$1A = \frac{1C}{1s}$$

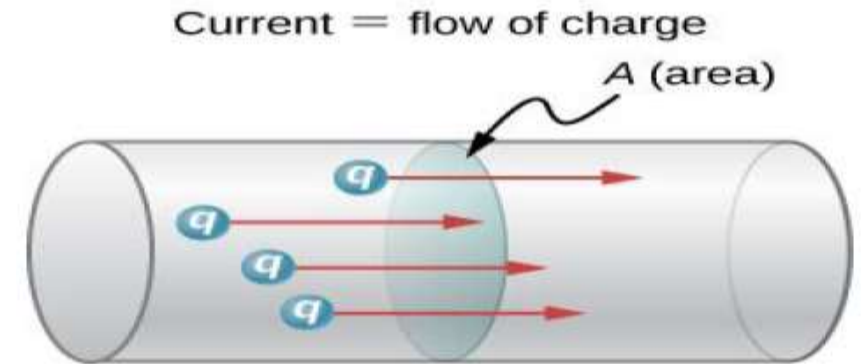


Fig. Current flow in a conductor

Ohm's Law

- states that “the voltage applied across the end points of a conductor is proportional to the flow of electric current.”

$$V \propto I \Rightarrow \frac{V}{I} = \text{Constant} = \text{Resistance (R)}$$

$$R = \frac{V}{I} \quad \text{or} \quad V = IR \quad \text{or} \quad I = \frac{V}{R} \quad (\text{Ohm's law})$$

- SI unit of R = Ohm(Ω): $1 \Omega = \frac{1V}{1A}$

- Ohm's law also stated as: “the current density $\vec{J}$ is proportional to the electric field $\vec{E}$.”

$$\vec{J} = \sigma \vec{E} \quad (\text{Ohm's law})$$

Where, σ is called electrical conductivity, and is a measure of a material's ability to conduct or transmit electricity.

Note: 1. Conductors have a higher electrical conductivity than insulators.

2. SI unit of σ is:

$$\sigma = \frac{[J]}{E} = \frac{A/m^2}{V/m} = \frac{A}{Vm} = \frac{1}{\Omega \cdot m}$$

Resistivity (ρ)

- is a measure of how strongly a material opposes the flow of electrical current.
- is the reciprocal of electrical conductivity.

$$\rho = \frac{1}{\sigma}$$

- SI unit of Resistivity is the Ohm-meter ($\Omega \cdot m$).
- Resistivity can be also defined in terms the electrical field and the current density:

$$\rho = \frac{\vec{E}}{\vec{J}} .$$

Electrical Energy

- the electric force on a charge q placed in an electric field E is given by

$$F = Eq$$

- the work W required to move a charge Q in a given electric field E

$$W = Fl, \quad \text{where } l \text{ is the length of the conductor}$$

$$W = Fl = (QE)l = Q(V/l)l = QV = U$$

Hence, the power is

$$P = \frac{U}{t} = \frac{QV}{t} = \frac{Q}{t}V = IV = I^2R = \frac{V^2}{R}$$

Equivalent Resistance and Kirchhoff's Rule

1. Resistors in Series

- $I = \text{constant.}$ $I_1 = I_2 = \dots = I_T$
- $V_T = \text{sum of each voltages.}$ $V_T = V_1 + V_2 + \dots V_N$

If N resistors are connected in series, the equivalent resistance is

$$R_{eq} = R_1 + R_2 + R_3 + \dots R_N = \sum_{i=1}^N R_i$$

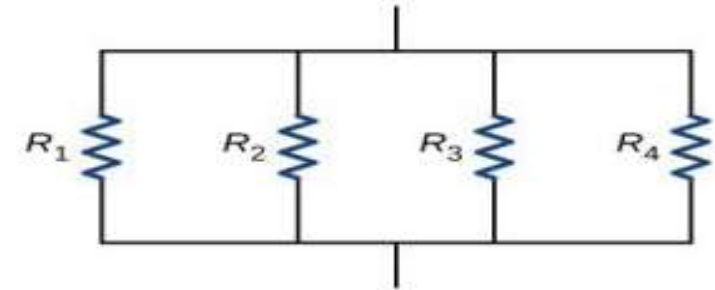


(a) Resistors connected in series

2. Resistors in Parallel

- $I_T = \text{sum of each currents :}$ $I_1 + I_2 + \dots + I_N$
- $V = \text{constant.}$ $V_T = V_1 = V_2 = \dots = V_N$
- the equivalent resistance R_{eq}

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \frac{1}{R_N} = \sum_{i=1}^N \left(\frac{1}{R_i} \right)$$



(b) Resistors connected in parallel

Kirchhoff's Circuit Rule

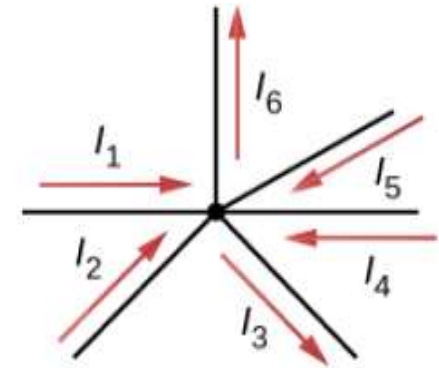
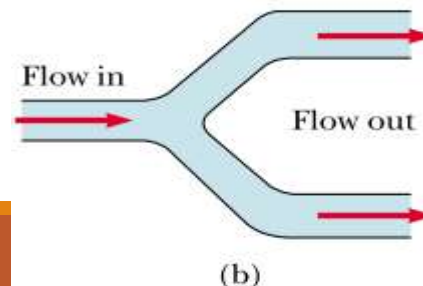
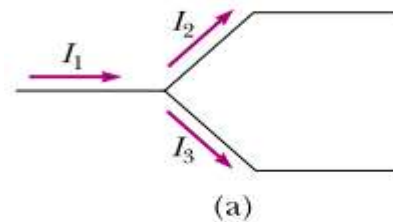
- Kirchhoff's Circuit Laws allow us to solve complex circuit problems by defining a set of basic network laws and theorems for the voltages and currents around a circuit.

1. Kirchhoff's First Rule (Junction Rule)

- states that the "total current or charge entering a junction or node is exactly equal to the charge leaving the node as it has no other place to go except to leave, as no charge is lost within the node"
- is also called Kirchhoff's Current Law (KCL)

In other words the algebraic sum of ALL the currents entering and leaving a node must be equal to zero, $I_{(\text{exiting})} + I_{(\text{entering})} = 0$. This idea by Kirchhoff is commonly known as the **Conservation of Charge**.

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$



$$\sum I_{\text{in}} = \sum I_{\text{out}} \\ I_1 + I_2 + I_4 + I_5 = I_3 + I_6$$

Currents entering a junction and exiting a junction

2. Kirchhoff's second rule (loop rule/Kirchhoff's Voltage Law(KVL))

States that “in any closed loop network, the total voltage around the loop is equal to the sum of all the voltage drops within the same loop” which is also equal to zero. In other words the algebraic sum of all voltages within the loop must be equal to zero. This idea by Kirchhoff is known as the **Conservation of Energy**.

The algebraic sum of changes in potential around any closed circuit path (loop) must be zero:

$$\Sigma V = 0.$$

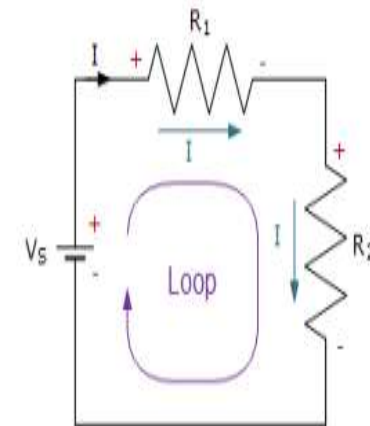
N:B. Sign convention

A. In emf(ϵ):

- if the direction of the **loop**, moves from **-ve to +ve** terminal of emf, then the voltage will be **+ve $\Rightarrow +\epsilon$**
- if the direction of the **loop**, moves from **+ve to -ve** terminal of emf, then the voltage will be **-ve $\Rightarrow -\epsilon$**

B. In the external Resistors (R):

- if the **direction of the loop** and the direction of **current I** are:
 - opposite $\Rightarrow$ the voltage (V) = **+ IR** (Positive)
 - the same $\Rightarrow$ the voltage (V) = **- IR** (negative)



Kirchhoff's loop consisting of two resistors and a voltage source

C. Potential difference between two points ($V_{ab} = V_a - V_b$):

- Start from point 'b' and move back to 'a', then

i) for emf (ϵ):

- if your direction of motion is from **-ve terminal of ϵ** to the **+ve terminal of ϵ** , then the voltage will be **+ve**.
 $\Rightarrow \text{emf} = +\epsilon$
- if your direction of motion is from **+ve terminal of ϵ** to the **-ve terminal of ϵ** , then the voltage will be **-ve**.
 $\Rightarrow \text{emf} = -\epsilon$

ii) In the external resistors (R):

- if **your direction of motion** & direction of **current I** are **opposite**, then the voltage will be **+ve**.
 $\Rightarrow V = +IR$
- if **your direction of motion** & direction of **current I** are **same**, then the voltage will be **-ve**.
 $\Rightarrow V = -IR$

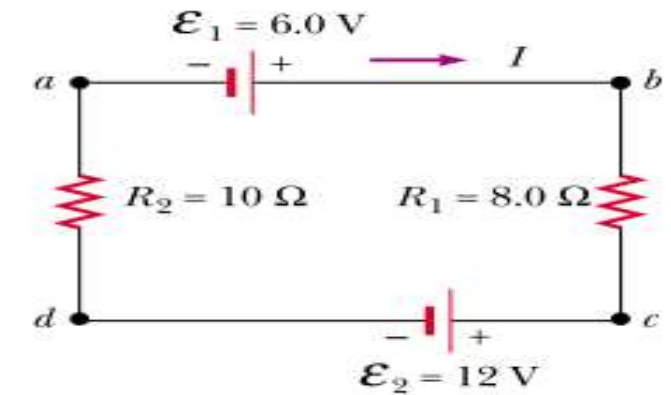
Note:

- Any capacitor acts as an open branch in a circuit. i.e the **current** in the branch containing the **capacitor** is **zero** under the steady-state condition.

Example-1

1. A single-loop circuit contains two resistors and two batteries, as shown in Fig

- (A) Find the current in the circuit.
- (B) What power is delivered to each resistor?
- (C) What power is delivered by the 12-V battery?



Solution:

A) Assume the direction of the loop is Clockwise:

$$\sum \Delta V = 0$$
$$\mathcal{E}_1 - IR_1 - \mathcal{E}_2 - IR_2 = 0 \quad \Rightarrow \quad I = \frac{\mathcal{E}_1 - \mathcal{E}_2}{R_1 + R_2} = \frac{6.0 \text{ V} - 12 \text{ V}}{8.0 \Omega + 10 \Omega} = -0.33 \text{ A}$$

B) Power by Resistors:

$$\mathcal{P}_1 = I^2 R_1 = (0.33 \text{ A})^2 (8.0 \Omega) = 0.87 \text{ W}$$

$$\mathcal{P}_2 = I^2 R_2 = (0.33 \text{ A})^2 (10 \Omega) = 1.1 \text{ W}$$

Hence, the total power delivered to the resistors is

$$\mathcal{P}_1 + \mathcal{P}_2 = 2.0 \text{ W.}$$

B) Power by emf 2

The 12-V battery delivers power $I\mathcal{E}_2 = 4.0 \text{ W}$.

Example-2

Find the currents I_1 , I_2 , and I_3 in the circuit shown in Fig below.

Solution. Assume clockwise loop:

$$(1) \quad I_1 + I_2 = I_3$$

$$(2) \quad \text{abcda} \quad 10.0 \text{ V} - (6.0 \, \Omega)I_1 - (2.0 \, \Omega)I_3 = 0$$

$$(3) \quad \text{befcb} \quad -14.0 \text{ V} + (6.0 \, \Omega)I_1 - 10.0 \text{ V} - (4.0 \, \Omega)I_2 = 0$$

Substituting Equation (1) into Equation (2)

gives

$$10.0 \text{ V} - (6.0 \, \Omega)I_1 - (2.0 \, \Omega)(I_1 + I_2) = 0$$

$$(4) \quad 10.0 \text{ V} = (8.0 \, \Omega)I_1 + (2.0 \, \Omega)I_2$$

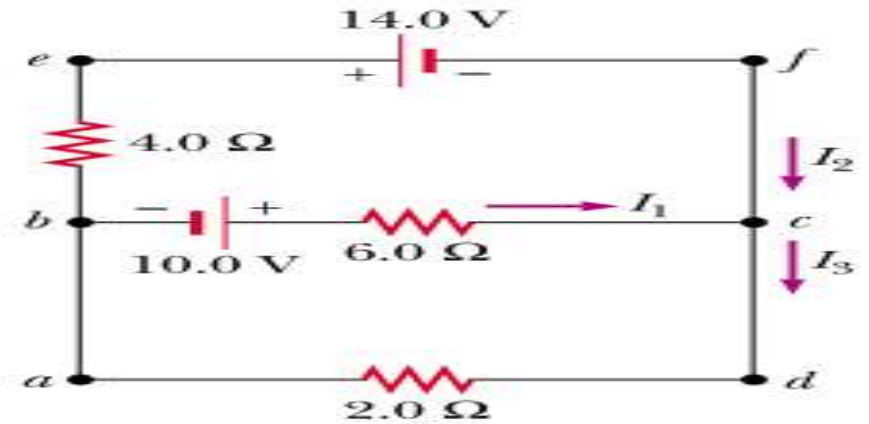
Dividing each term in Equation (3) by 2 and rearranging gives

$$(5) \quad -12.0 \text{ V} = -(3.0 \, \Omega)I_1 + (2.0 \, \Omega)I_2$$

Subtracting Equation (5) from Equation (4) eliminates I_2 , giving

$$22.0 \text{ V} = (11.0 \, \Omega)I_1$$

$$I_1 = 2.0 \text{ A}$$



Using this value of I_1 in Equation (5) gives a value for I_2 :

$$\begin{aligned} (2.0 \, \Omega)I_2 &= (3.0 \, \Omega)I_1 - 12.0 \text{ V} \\ &= (3.0 \, \Omega)(2.0 \text{ A}) - 12.0 \text{ V} = -6.0 \text{ V} \end{aligned}$$

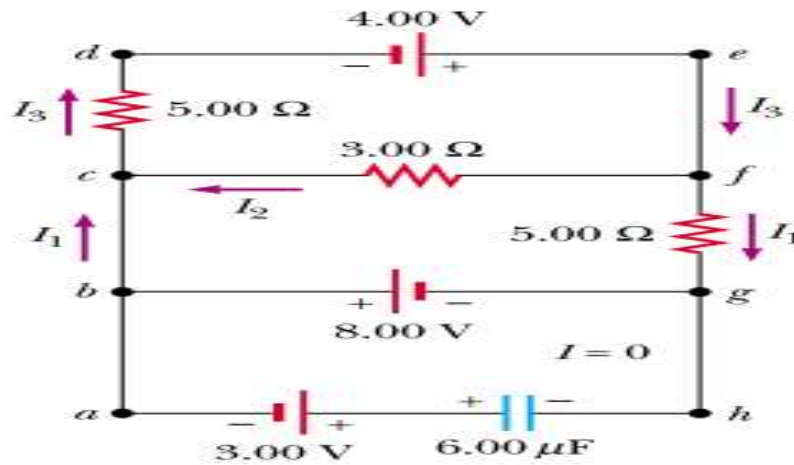
$$I_2 = -3.0 \text{ A}$$

Finally,

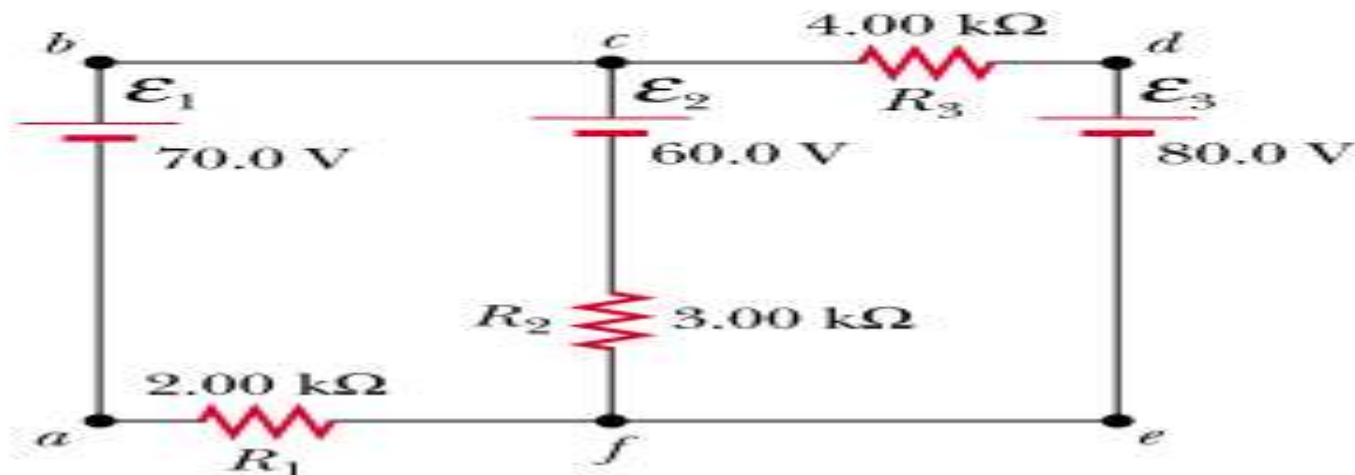
$$I_3 = I_1 + I_2 = -1.0 \text{ A}$$

Homework

- (A) Under steady-state conditions, find the unknown currents I_1 , I_2 , and I_3 in the multiloop circuit shown in Fig below.
(B) What is the charge on the capacitor?



2. Using Kirchhoff's rules, (a) find the current in each resistor in Figure (b) Find the potential difference between points c and f . Which point is at the higher potential?



20. The ammeter shown in Figure P28.20 reads 2.00 A. Find I_1 , I_2 , and $\mathcal{E}$.

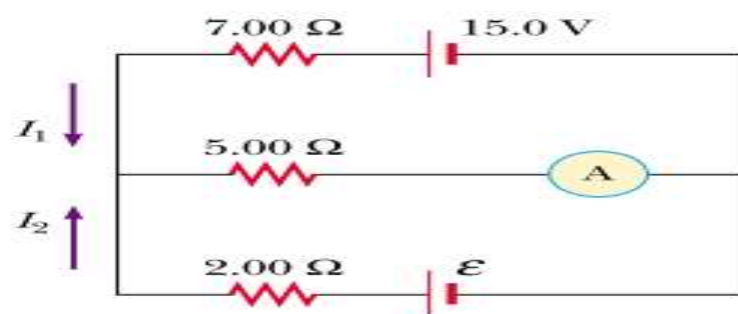
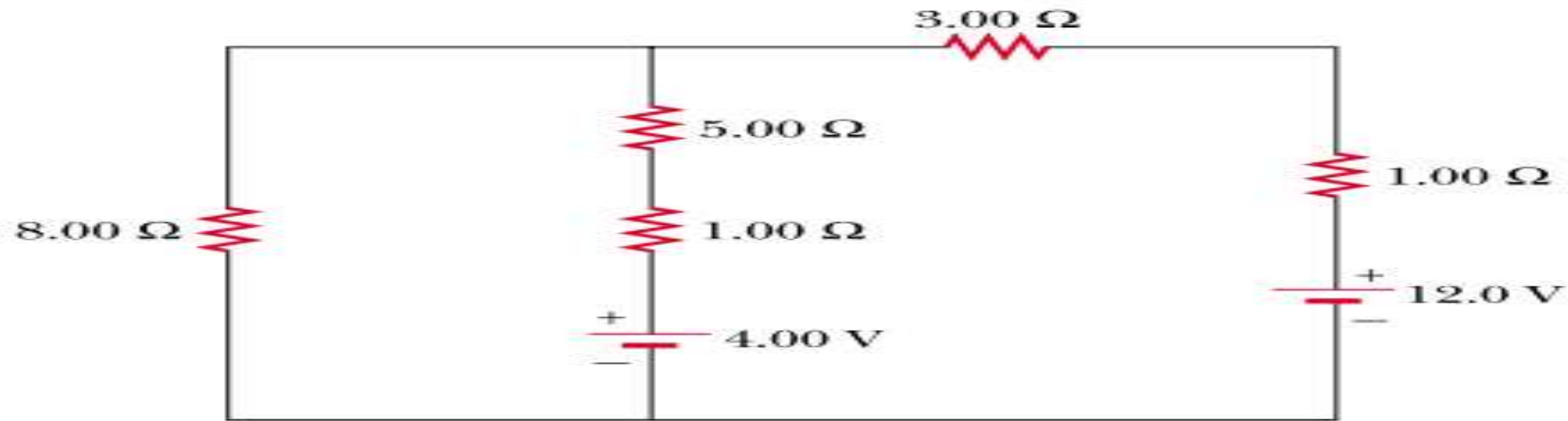


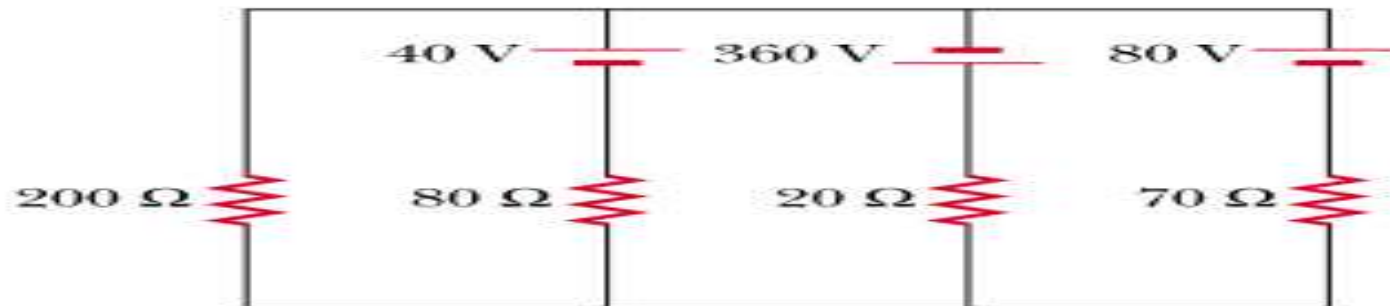
Figure P28.20

21. 

Determine the current in each branch of the circuit shown in Figure P28.21.



26. In the circuit of Figure P28.26, determine the current in each resistor and the voltage across the 200-Ω resistor.



27. A dead battery is charged by connecting it to the live battery of another car with jumper cables (Fig. P28.27). Determine the current in the starter and in the dead battery.

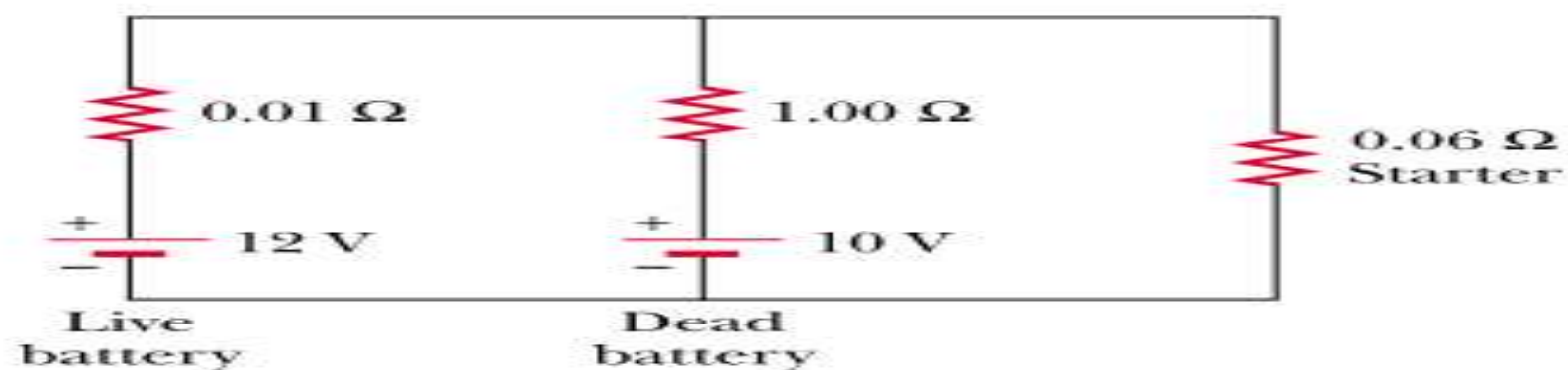


Figure P28.27

Magnetic Field and Magnetic Flux

Definition: Magnetic Field

- **A magnetic** field is defined by the force that a charged particle experiences moving in this field, after we account for the gravitational and any additional electric forces possible on the charge.
- is a region where magnetic force is felt
- The magnitude of this force is proportional to the amount of charge q , the speed of the charged particle v , and the magnitude of the applied magnetic field. And is given by:

$$\vec{F} = q\vec{V} \times \vec{B} \quad \Rightarrow \quad F = qvB\sin\theta$$

- ✓ Where, θ is the angle between the velocity and the magnetic field.
- ✓ The SI unit for magnetic field strength B is called the tesla (T)

A smaller unit, called the gauss (G), where $1\text{G} = 1 \times 10^{-4}\text{T}$, is sometimes used.

Note: The direction of this force is perpendicular to both the direction of the moving charged particle and the direction of the applied magnetic field.

The magnetic flux (Φ)

- can be visualized as the number of magnetic field lines passing through a given area.
- The number of magnetic field lines per unit area, i.e. B , is then referred to as the density of the magnetic flux or, more properly, the magnetic flux density.

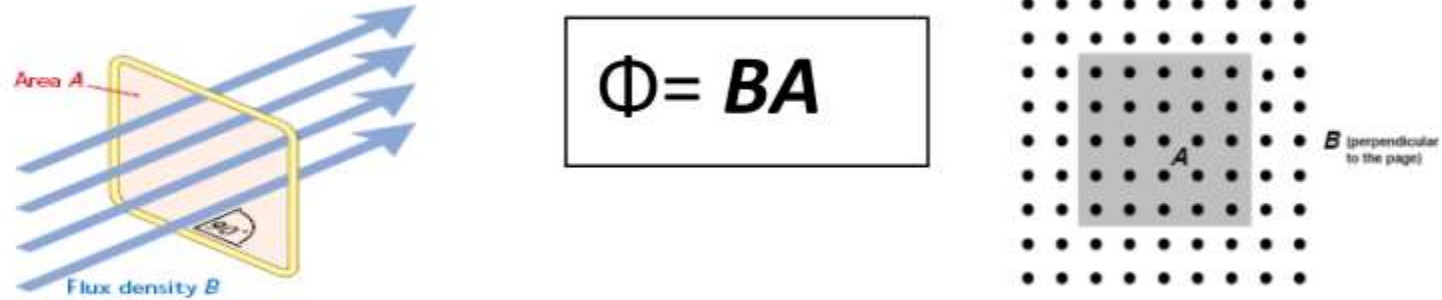


Figure **Magnetic flux through a loop of area A**

Electromagnetic Induction

- Electromagnetic induction occurs when an emf is induced in a coil due to a changing magnetic flux. This is known as Faraday's Law of Electromagnetic Induction.
- Movement of a magnet relative to a coil produces emfs. The same emfs are produced if the coil is moved relative to the magnet
- Electromotive force of opposite signs are produced by motion in opposite directions and the directions of emfs are also reversed by reversing poles. The same results are produced if the coil is moved rather than the magnet; it is the relative motion that is important.

The Laws of Electromagnetic Induction

1. **Faraday's Law of Induction:** states that the size of the emf induced in a circuit is directly proportional to the rate of change of magnetic flux.
2. **Lenz's Law states:** that the direction of the induced emf is always such as to oppose the change producing it.

$$\text{Induced emf} = -N \frac{\Delta \Phi}{\Delta t}$$

Where N = the number of turns

N:B. the negative sign is due to Lenz's law

- Lenz's law states that "The induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop."

Example

The square coil of (Figure) has sides $l=0.2\text{m}$ long and is tightly wound with $N=200$ turns of wire. The resistance of the coil is $R=5.0\Omega$. The coil is placed in a spatially uniform magnetic field that is directed perpendicular to the face of the coil and whose magnitude is decreasing at a rate $\frac{\Delta B}{\Delta t} = -0.04\text{T/s}$. (a) What is the magnitude of the emf induced in the coil? (b) What is the magnitude of the current circulating through the coil?

Solution:

a) The flux through the square loop is given by

$$\Phi = \vec{B} \cdot \vec{A} = Bl^2$$

Then the induced emf from Faraday's law is,

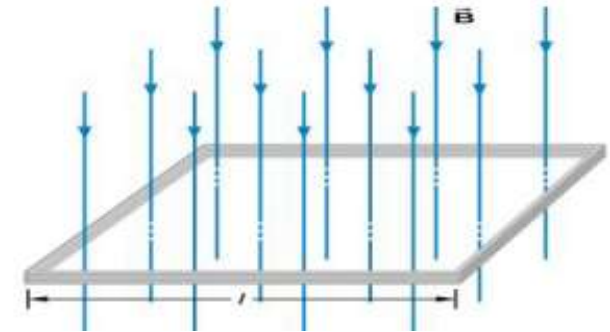
$$\varepsilon = N \frac{\Delta \Phi}{\Delta t} = NA \frac{\Delta B}{\Delta t} = (200)(0.2\text{m})^2 (0.04\text{T/s})$$

$$\varepsilon = 0.5\text{V}$$

b) Then the current can be calculated from Ohm's law

$$\varepsilon = IR$$

$$I = \frac{\varepsilon}{R} = \frac{0.5\text{V}}{5\Omega} = 0.1\text{A}$$



The End

(July 19, 2022)